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Coordinated planning and operation of SST-based DC data centers with tiered workload flexibility☆

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HIGHLIGHTS

- Unified planning coordinates power and computing in SST-based DC data centers.
- SST losses across operating points are embedded in long-term capacity planning.
- SLO-tiered workload flexibility reshapes server and power system sizing.
- Enhanced Benders decomposition solves the annual multi-scenario mixed-integer model.

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ABSTRACT

The rapid growth of artificial intelligence (AI) workloads is making modern data centers increasingly energy-intensive and reliability-critical. This trend not only raises electricity expenditure but also amplifies the economic impact of power interruptions, motivating coordinated planning of power infrastructure and computing resources. Against this background, solid-state transformer (SST)-based DC supply architectures are emerging as a promising direction for next-generation data centers because they provide a controllable interface for integrating diverse energy and computing resources. Under this architectural setting, this paper proposes a coordinated planning and operation framework for SST-based DC data centers with tiered workload flexibility. The framework incorporates nonlinear SST losses into long-term capacity planning, treats tiered workload flexibility as a planning resource, and coordinates energy storage and diesel generators for backup supply during utility outages. An annual bi-level model is formulated to jointly determine the capacities of SST conversion stages, renewable generation, energy storage, diesel generators, and heterogeneous computing servers, together with their scenario-dependent operating decisions. Piecewise linear approximation is used to reformulate the SST loss model, and an enhanced Benders decomposition algorithm is developed to solve the resulting multi-scenario mixed-integer problem. Case study results demonstrate that the proposed framework reduces total annual cost while satisfying workload service constraints. Further analysis shows that nonlinear SST losses influence capacity configuration, whereas tiered workload flexibility reshapes server deployment and backup resource requirements.

1. Introduction

1.1. Research background

Driven by the rapid development of artificial intelligence (AI), foundation-model training and inference, and cloud computing services, modern data centers are experiencing unprecedented growth in computing demand [1,2]. Recent analyses indicate that global electricity demand from data centers is projected to more than double by 2030 and reach approximately 945 TWh, with AI deployment serving as a major driver of this increase [3,4]. Against this background, high-density computing clusters and GPU-intensive workloads are becoming increasingly prevalent, making data centers one of the most electricity-intensive and reliability-critical load types in modern power systems. The increasing electricity consumption of data centers substantially raises operating expenditure, while stringent service-continuity requirements make power supply reliability a fundamental

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Nomenclature

Sets and indices

s	Operating scenario index from 1 to S
t	Time interval index from 1 to T
n	Node index from 1 to N
i,j	DC network node indices in Ω_{dc}
ij	Directed branch index in L_{dc}
k	Server type index from 1 to K
ρ	Workload type index from 1 to Q
m	Front-end server index from 1 to M
l	Computing frequency level index from 1 to L
$\widehat{t}$	Deadline time interval
r	Breakpoint index from 1 to R
g	SST conversion stage index
α	Auxiliary variable index from 1 to $B_{n,k}$
ν	Iteration index

Parameters

$\alpha_{\mathrm{AD}}^{\mathrm{SST}},\beta_{\mathrm{AD}}^{\mathrm{SST}},\gamma_{\mathrm{AD}}^{\mathrm{SST}}$	Loss coefficients of the AC/DC stage
$\alpha_{\mathrm{DD}}^{\mathrm{SST}},\beta_{\mathrm{DD}}^{\mathrm{SST}},\gamma_{\mathrm{DD}}^{\mathrm{SST}}$	Loss coefficients of the DC/DC stage
$\lambda_{pri},\lambda_{sec}$	Current conversion coefficients
$P_{n,k}^{\mathrm{fix}}$	Fixed power of a type-k server
κ_1	Power consumption coefficient
κ_2	Computing efficiency coefficient
$\omega_{k,\rho}$	Relative processing capability coefficient
$f_{n,k,l}^{\mathrm{cmp}}$	Computing frequency at level l
$N_{n,k}^{\mathrm{mod}}$	Number of servers in one unit module
η^{PUE}	Power usage effectiveness of the data center
τ_{ρ}^{max}	Maximum tolerable delay of type-ρ workload
$\tau_{\rho}^{\mathrm{tol}}$	Delay tolerance of type-ρ workload
$q_n^{\max}$	Maximum stored workload at node n
β^{sec}	Secure operating factor of computing resources
$\eta^{\mathrm{ch}},\eta^{\mathrm{dis}}$	Charging and discharging efficiencies of ESS
$\overline{P}_n^{\mathrm{ch}},\overline{P}_n^{\mathrm{dis}}$	Charging and discharging power limits of ESS
$\overline{P}_{s,n,t}^{\mathrm{PV}},\overline{P}_{s,n,t}^{\mathrm{WT}}$	Available factor of PV and WT
$\overline{P}_n^{\mathrm{DG}},\overline{P}_n^{\mathrm{DG}}$	Minimum and maximum output limits
$R_n^{\mathrm{DG,up}},R_n^{\mathrm{DG,down}}$	Ramp-up and ramp-down limit of DG
$T_{\min}^{\mathrm{DG}}$	Minimum continuous operating duration of DG
r_{ij}	Equivalent resistance of branch ij
$\overline{P}_{ij}$	Maximum allowable active power flow of branch ij
$\overline{V}_n,\overline{V}_n$	Lower and upper voltage limits of node n
$\delta_{s,t}^{\mathrm{UG}}$	Utility power supply availability parameter
$\overline{P}_n^{\mathrm{UG}}$	Maximum utility power injected at node n
δ	Discount rate of equipment
ℓ	Equipment lifetime
$S_{\mathrm{AD,mod}}^{\mathrm{SST}},S_{\mathrm{DD,mod}}^{\mathrm{SST}}$	Unit capacity of the SST conversion stage
$S_{\mathrm{mod}}^{\mathrm{PV}},S_{\mathrm{mod}}^{\mathrm{WT}},S_{\mathrm{mod}}^{\mathrm{DG}}$	Unit capacity of PV, WT and DG
$\overline{S}_n^{\mathrm{PV}},\overline{S}_n^{\mathrm{WT}},\overline{S}_n^{\mathrm{DG}}$	Upper bounds of capacities
$E_{\mathrm{mod}}^{\mathrm{ESS}}$	Unit capacity of ESS
$\overline{E}_n^{\mathrm{ESS}}$	Upper bounds of ESS capacity
$c_{\mathrm{ALC}}^{\mathrm{SST}},c_{\mathrm{ALC}}^{\mathrm{ESS}},c_{\mathrm{ALC}}^{\mathrm{PV}},c_{\mathrm{ALC}}^{\mathrm{WT}},c_{\mathrm{ALC}}^{\mathrm{DG}}$	Unit investment cost coefficients of SST, ESS, PV, WT and DG
$c_{\mathrm{ALC},k}^{\mathrm{srv}}$	Unit investment cost coefficient of type-k servers
$c_{\mathrm{OPE}}^{\mathrm{SST}}$	Unit operation and maintenance cost coefficient
c_t^{ESS}	ESS operating cost coefficient
$c_{\mathrm{curt}}^{\mathrm{PV}},c_{\mathrm{curt}}^{\mathrm{WT}}$	Curtailment penalty coefficient of PV and WT
c_t^{DG}	Unit operating cost coefficient of the DG
c_t^{UG}	Utility electricity purchasing price
c^L	Load curtailment cost coefficient
p_s	Probability of each scenario

Variables

$S_{\mathrm{AD}}^{\mathrm{SST}},S_{\mathrm{DD}}^{\mathrm{SST}}$	Planned capacities of the AC/DC and DC/DC stages
$k_{\mathrm{AD}}^{\mathrm{SST}},k_{\mathrm{DD}}^{\mathrm{SST}}$	Numbers of AC/DC and DC/DC stage modules
E_n^{ESS}	Planned ESS capacity at node n
k_n^{ESS}	Number of ESS modules at node n
$S_n^{\mathrm{PV}},S_n^{\mathrm{WT}},S_n^{\mathrm{DG}}$	Planned capacities of PV, WT, and DG at node n
$k_n^{\mathrm{PV}},k_n^{\mathrm{WT}},k_n^{\mathrm{DG}}$	Numbers of PV, WT, and DG modules at node n
$z_{n,k,\alpha}$	Binary variable for type-$k$server deployment
$N_{n,k}^{\mathrm{srv}}$	Number of type-k servers deployed at node n
$P_{t,a}^{\mathrm{SST}},P_{t,b}^{\mathrm{SST}},P_{t,c}^{\mathrm{SST}}$	SST power at different interface ports
$P_{\mathrm{AD},t}^{\mathrm{SST}},P_{\mathrm{DD},t}^{\mathrm{SST}}$	Active powers transferred through the SST conversion stages in time interval t
$I_{\mathrm{AD},t}^{\mathrm{SST}},I_{\mathrm{DD},t}^{\mathrm{SST}}$	Equivalent currents of the SST conversion stages in time interval t
$P_{\mathrm{AD},t}^{\mathrm{SST,loss}},P_{\mathrm{DD},t}^{\mathrm{SST,loss}}$	Power losses of the AC/DC and DC/DC stages in time interval t
$P_{s,n,t}^{\mathrm{IT}}$	IT power demand at node n in time interval t under scenario s
$P_{s,n,t}^{\mathrm{aux}}$	Auxiliary power demand at node n in time interval t under scenario s
$P_{s,n,t}^{\mathrm{tot}}$	Total data center power demand at node n in time interval t under scenario s
$P_{s,n,k,t}^{\mathrm{srv}}$	Power consumption of type-k servers at node n in time interval t under scenario s
$P_{s,n,k,t}^{\mathrm{cmp}}$	Computing power of type-k servers at node n in time interval t under scenario s
$u_{s,n,k,t,l}$	Binary variable for selecting frequency level l of type-k servers at node n in time interval $t$under scenario s
$f_{s,n,k,t}^{\mathrm{cmp}}$	Selected computing frequency of type-k servers at node n in time interval t under scenario s
$\varphi_{s,n,k,t}$	Service capacity of one type-k servers at node n in time interval t under scenario s
$H_{s,n,k,t}^{\mathrm{cmp}}$	Total service capacity of type-k servers at node n in time interval t under scenario s
$v_{s,n,k,t}^{\mathrm{srv}}$	Service rate of type-k servers at node n in time interval t under scenario s
$a_{s,n,\rho,m,t}^{\mathrm{cmp}}$	Workload of type-ρ assigned from front-end server m to node n in time interval t under scenario s
$p_{s,n,\rho,t}^{\mathrm{cmp}}$	Processed workload of type-ρ at node n in time interval t under scenario s
$p_{s,n,k,\rho,t}^{\mathrm{cmp}}$	Workload of type-ρ processed by type-k servers at node n in time interval t under scenario s
$\Delta q_{s,n,\rho,t}^{\mathrm{cmp}}$	Change in stored workload of type-ρ at node n in time interval t under scenario s
$q_{s,n,\rho,t}^{\mathrm{cmp}}$	Stored workload of type-ρ at node n in time interval t under scenario s
$A_{s,\rho,\widehat{t}}^{\mathrm{req}}$	Required cumulative amount of type-ρ delay-tolerant workload by deadline $\widehat{t}$
$E_{s,n,t}^{\mathrm{ESS}}$	Stored energy of the ESS at node n in time interval t under scenario s
$P_{s,n,t}^{\mathrm{ch}},P_{s,n,t}^{\mathrm{dis}}$	Charging and discharging powers of the ESS at node n in time interval t under scenario s
$u_{s,n,t}^{\mathrm{ch}},u_{s,n,t}^{\mathrm{dis}}$	Binary variables for ESS charging and discharging states at node n in time interval t under scenario s
$P_{s,n,t}^{\mathrm{PV}},P_{s,n,t}^{\mathrm{WT}}$	PV and WT power outputs at node n in time interval t under scenario s

$P^{DG}_{s,n,t}$	DG output power at node n in time interval t under scenario s	$P_{s,n,t}$	Net power injection at node n in time interval t under scenario s
$y^{DG}_{s,n,t}$	Binary variable for DG operating state at node n in time interval t under scenario s	$V_{s,n,t}$	Voltage magnitude at node n in time interval t under scenario s
$u^{DG}_{s,n,t}, v^{DG}_{s,n,t}$	Binary variables for DG startup and shutdown states at node n in time interval t under scenario s	$\Delta P^{shed}_{s,n,t}$	Load shedding at node n in time interval t under scenario s
$P^{UG}_{s,n,t}$	Utility power injected at node n in time interval t under scenario s	C_{ALC}	Annualized investment cost
$P_{s,ij,t}$	Power flow on branch ij in time interval t under scenario s	C_{OPE}	Expected operating cost
		C_{REL}	Reliability cost

design consideration. Power interruptions may lead to service outages, data loss, and significant economic losses [5]. As a consequence, coordinated planning of power infrastructure and computing resources has become increasingly important for next-generation data centers [6].

1.2. Related work

From a planning perspective, the evolution of power supply architectures directly affects data center efficiency, resource integration, and long-term capacity allocation [7]. Conventional data center power supply systems are mainly based on multi-stage AC conversion chains, where utility power is processed through several conversion and distribution stages before reaching information technology (IT) equipment. Such architectures introduce additional losses and limit the direct integration of renewable generation and storage resources. In contrast, DC distribution architectures have attracted increasing attention because they can reduce redundant conversion stages and facilitate the integration of renewable energy and battery storage [8,9]. Among the enabling technologies for such architectures, the solid-state transformer (SST) is particularly promising because it can support flexible power conversion among utility-side AC systems, medium-voltage DC buses, and low-voltage DC distribution networks, while also enabling controllable power routing and fast dynamic response [10,11]. These characteristics make SST-centered DC supply architectures a promising option for data centers that require coordinated management of electricity supply, local resources, and computing demand.

In addition to power supply architecture, workload characteristics have become another important factor in data center planning. Modern data center loads can no longer be adequately represented by a fixed aggregate demand profile. Owing to the increasing diversification of computing services, different workloads exhibit heterogeneous delay requirements, service priorities, and allowable delay windows. In particular, delay-sensitive workloads must be processed immediately to satisfy stringent quality of service constraints, whereas a growing share of delay-tolerant workloads can be scheduled more flexibly over time [12,13] or operated at different performance levels [14]. Such workload heterogeneity creates intrinsic demand-side flexibility and calls for coordinated management of computing resources and power consumption [15–17]. Recent studies have demonstrated data center workload flexibility in batch processing, renewable-aware operation, dynamic overbooking, frequency regulation, and grid-interactive demand response [18–21]. Recent demonstrations further indicate that AI data center loads can provide dispatchable demand response under differentiated service constraints, achieving a sustained 25% power reduction for 3 h during peak-demand periods while maintaining quality of service compliance [22]. Therefore, treating IT demand as a fixed exogenous power curve removes an important source of planning flexibility [23–25]. A more realistic formulation should capture workload categories, server heterogeneity, and service-related operating constraints, so that computing-side flexibility can be coordinated with power-side resource planning [26,27].

Moreover, long-term data center planning should consider not only normal operation but also investment decisions and service continuity during utility outages. Prior work has shown that flexible computing demand, renewable generation, and energy storage can materially affect data center energy planning [28]. Backup resources remain essential for maintaining service continuity under such conditions. Diesel generators can provide sustained backup power during prolonged outages, but their configuration involves a tradeoff between reliability and economic cost [29]. The role of backup resources has also been examined from the perspectives of data center reliability and microgrid resilience [29,30]. Taken together, these studies suggest that modern data center planning requires closer coordination among power supply architecture, workload flexibility, local energy resources, and backup resource configuration [31–33].

1.3. Research gaps

Previous studies on data center power architectures, workload flexibility, renewable integration, and energy management provide useful foundations for improving data center energy performance. However, several gaps remain in applying these advances to long-term planning of SST-based DC data centers.

First, the annual planning implications of SST-based DC supply architectures and nonlinear conversion losses have not been sufficiently quantified. Existing SST studies have mainly focused on SST-enabled DC microgrid integration and steady-state power flow modeling [34–36]. These studies provide useful foundations for system integration and steady-state analysis, but they rarely examine how SST conversion capacity and nonlinear losses affect the long-term sizing of data center power infrastructure and computing resources. Meanwhile, existing data center planning studies are still largely formulated for conventional AC power supply architectures or represent the power supply side through conventional microgrid structures and aggregate grid interfaces [28,37]. Dedicated long-term planning studies for SST-based DC data centers therefore remain limited. Accordingly, the influence of SST conversion characteristics on the coordinated sizing of power and computing resources remains insufficiently examined.

Second, workload flexibility has not been sufficiently treated as a long-term planning resource in SST-based DC data centers. Many existing planning studies represent computing demand as an exogenous power profile and size power resources according to the resulting load curve. This treatment separates computing adaptability from infrastructure planning and overlooks the ability of workload scheduling to reshape temporal power demand. Although recent studies have incorporated data center flexibility into long-term planning [38,39], they rarely link it with the sizing of SST conversion stages and associated power resources. In addition, existing workload models often consider only batch workload shifting [37] or a simple distinction between delay-sensitive and delay-tolerant services [40]. Such formulations may not fully represent the tiered flexibility arising from different service requirements, performance levels, and delay tolerances. Therefore, tiered workload flexibility should be incorporated into long-term planning to quantify its effects on server deployment and power infrastructure sizing.

Third, the complementary roles of energy storage systems and diesel

generators during utility outages have not been sufficiently represented in long-term data center planning. Existing studies often focus primarily on storage-based backup [41,42], making it difficult to distinguish the capacity requirements of different backup technologies under varying outage durations. Energy storage systems are generally suitable for rapid support during short interruptions, whereas diesel generators can provide sustained supply during prolonged outages. Battery storage has been investigated for improving data center reliability, while the coordinated use of storage and diesel generation has also been studied in microgrid resilience [29,30]. However, their joint sizing has rarely been considered together with SST-based DC architecture and workload flexibility. Neglecting this coordination may result in suboptimal backup capacity allocation and an incomplete assessment of the economic value of coordinated energy storage and diesel generation under different outage conditions [6,43].

To further clarify the position of this study, Table 1 compares the proposed framework with representative studies in the existing literature. Existing research has made important progress in SST-enabled DC systems, data center workload flexibility, energy resource planning, and backup resource configuration. However, these research areas have largely developed in parallel. Although SST-based DC supply architectures are emerging as a promising direction for next-generation data centers, their annual planning implications remain insufficiently explored. In particular, a dedicated planning formulation that connects SST conversion characteristics, computing flexibility, local energy resources, and backup resource configuration within a unified annual framework is still lacking. Table 1 highlights this gap and shows how the proposed framework connects these previously separate research directions.

1.4. Primary contributions

To address the identified research gaps, this study develops an annual coordinated planning and operation framework for SST-based DC data centers with tiered workload flexibility. The upper level determines the capacities of SST conversion stages, renewable generation, energy storage systems, diesel generators, and heterogeneous computing servers, while the lower level optimizes the operation of power and computing resources under normal conditions and utility outages. Nonlinear SST losses are reformulated using piecewise linear approximation, and an enhanced Benders decomposition method is employed to solve the resulting annual multi-scenario mixed-integer planning problem. The main contributions of this paper are summarized as follows.

1) A dedicated annual coordinated planning framework is established for SST-based DC data centers. The framework explicitly incorporates SST conversion capacity and loss characteristics into the coordinated planning of power and computing resources, enabling the long-term resource configuration, operating cost, and reliability-related cost of this emerging supply architecture to be quantified.
2) Tiered workload flexibility is modeled as a planning resource and linked with power infrastructure sizing. By considering workload scheduling under differentiated service requirements, dynamic voltage and frequency scaling (DVFS) operation, and heterogeneous server deployment, the model evaluates how computing flexibility reshapes server configuration, SST capacity, and backup resource requirements.
3) Normal operation and utility outages are represented in a multi-scenario formulation to coordinate the complementary backup roles of energy storage and diesel generators. Nonlinear SST losses are reformulated using piecewise linear approximation, and an enhanced Benders decomposition is adopted to solve the resulting large-scale mixed-integer problem efficiently.

The rest of this paper is organized as follows. Section 2 depicts the system modeling of the SST-based DC data center. Section 3 presents the coordinated multi-resource planning model. Section 4 develops the enhanced Benders decomposition method. Section 5 reports the case studies to verify the efficacy of the proposed method. Section 6 concludes the paper.

2. System modeling of the SST-based DC data center

2.1. DC data center architecture

With the increasing power density of AI data centers, DC supply architectures have attracted growing attention because they can reduce redundant conversion stages and facilitate the integration of renewable generation and energy storage. The DC data center studied in this paper adopts a 2 N power supply architecture to meet the reliability requirements of critical computing services, as shown in Fig. 1. Two independent utility supply paths and corresponding conversion branches are configured so that either path can support critical loads when the other is unavailable. In this representative architecture, two SSTs connect the utility-side AC system to the internal medium-voltage and low-voltage DC buses. These controllable interfaces enable the coordinated integration of utility supply, renewable generation, energy storage, diesel generators, and computing loads within a unified electrical framework.

Within the proposed architecture, the two utility supply paths are connected at nodes 1 and 2, while the diesel generators (DGs) are connected to the AC supply branches at nodes 3 and 4. For the left SST, the AC port, medium-voltage DC port, and low-voltage DC port correspond to nodes 3, 5, and 9, respectively. The corresponding ports of the right SST are nodes 4, 6, and 10. Photovoltaic (PV) generation and wind

Table 1
Comparison of the proposed work with relevant studies.

Ref	SST-based DC architecture	Long-term capacity planning	Workload flexibility	Multi-resource configuration				Utility-supply outage scenarios	Two-stage optimization
				PV	WT	DG	ESS		
[6]	✘	✔	✔	✔	✘	✘	✔	✔	✔
[20]	✘	✘	✔	✔	✔	✘	✘	✘	✘
[21]	✘	✘	✔	✘	✘	✘	✘	✘	✔
[28]	✘	✔	✔	✘	✘	✘	✔	✘	✘
[29]	✘	✔	✘	✔	✘	✔	✔	✔	✘
[30]	✘	✔	✘	✔	✔	✔	✔	✔	✘
[36]	✔	✘	✘	✔	✘	✘	✔	✔	✘
[37]	✘	✔	✔	✔	✔	✘	✔	✘	✔
[39]	✘	✔	✔	✘	✘	✘	✘	✘	✔
[40]	✘	✔	✔	✔	✔	✘	✔	✘	✔
This study	✔	✔	✔	✔	✔	✔	✔	✔	✔

PV – photovoltaic generation; WT – wind turbine; DG – diesel generator; ESS – energy storage system.

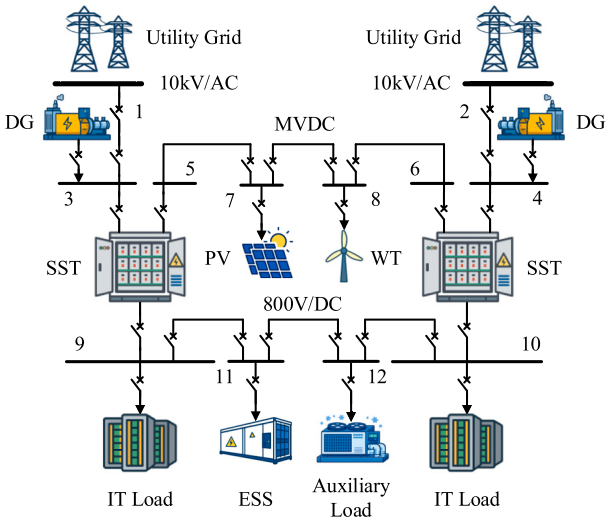


Fig. 1. System architecture of the SST-based DC data center with integrated multi-energy resources.

turbines (WTs) are connected to the medium-voltage DC side at nodes 7 and 8, respectively. The energy storage system (ESS) is connected at node 11, the auxiliary load at node 12, and the IT loads at nodes 9 and 10.

The utility and DG branches on the AC side are not included as ordinary nodes in the internal DC branch-flow model. Instead, their active power is transferred into the DC network through the corresponding SST conversion stages. Consequently, SST conversion capacity and losses are coupled with renewable generation, ESS operation, DG backup support, and computing-side load scheduling in the planning model. This architecture provides the physical basis for the subsequent resource and network formulations.

2.2. SST power conversion and loss characterization

The SST serves as the key power conversion interface between the upstream utility grid and the internal DC distribution network of the data center. In the proposed architecture, the SST connects the 10 kV utility-side AC system, an internal medium-voltage DC (MVDC) bus, and the 800 V low-voltage DC distribution bus. Therefore, it provides the electrical interface for utility power access, DC power distribution, and multi-resource integration within the data center. In this paper, an active-power-based SST power flow model is developed to describe the stage-wise power transfer process through the AC/DC and DC/DC conversion stages. As shown in Fig. 2, the SST consists of an AC/DC conversion stage and a DC/DC conversion stage.

The conversion losses are generated in both the AC/DC and DC/DC stages. To explicitly capture their influence on power transfer and capacity planning, the losses of the two conversion stages are separately represented in the SST power flow model. The corresponding loss locations are illustrated in Fig. 3.

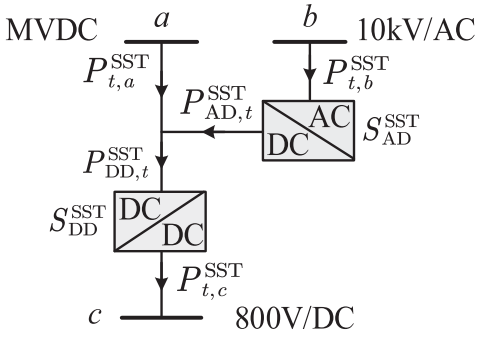


Fig. 2. Internal power conversion stages of the SST.

According to the power transfer process of the SST, the power balance of the AC/DC stage can be written as:

$$P^{SST}_{t,b} = P^{SST}_{AD,t} + P^{SST,loss}_{AD,t} \tag{1}$$

where $P^{SST}_{t,b}$ denotes the active power absorbed from the 10 kV/AC bus, $P^{SST}_{AD,t}$ denotes the dc-side output power of the AC/DC stage, and $P^{SST,loss}_{AD,t}$ denotes the corresponding conversion loss.

At the intermediate dc link, the power entering the DC/DC stage consists of the power imported from the MVDC bus and the dc-side output of the AC/DC stage:

$$P^{SST}_{DD,t} = P^{SST}_{t,a} + P^{SST}_{AD,t} \tag{2}$$

where $P^{SST}_{t,a}$ is the active power absorbed from the MVDC bus, and $P^{SST}_{DD,t}$ is the input power of the DC/DC stage.

The power balance of the DC/DC stage is further written as:

$$P^{SST}_{DD,t} = P^{SST}_{t,c} + P^{SST,loss}_{DD,t} \tag{3}$$

where $P^{SST}_{t,c}$ denotes the output power delivered to the 800 V/DC bus, and $P^{SST,loss}_{DD,t}$ denotes the conversion loss of the DC/DC stage.

Existing studies have shown that the conversion losses of SST power electronic stages are strongly dependent on converter loading. Fig. 4 presents a representative loss profile adopted from the literature [44] for the AC/DC and DC/DC conversion stages, showing the nonlinear relationship between converter current and power loss. As the converter current increases, the losses of both stages increase nonlinearly. In particular, the DC/DC stage exhibits a larger loss magnitude and a steeper growth trend over most of the operating range. Therefore, simplified fixed-efficiency representations may not accurately capture the load-dependent loss characteristics of SSTs in long-term planning studies.

To characterize this nonlinear relationship while maintaining model tractability, the loss of the AC/DC stage is modeled as a quadratic function of the equivalent converter current:

$$P^{SST}_{loss,AD,t} = \alpha^{SST}_{AD} + \beta^{SST}_{AD} I^{SST}_{AD,t} + \gamma^{SST}_{AD} \left(I^{SST}_{AD,t}\right)^2 \tag{4}$$

where α^{SST}_{AD}, β^{SST}_{AD} and γ^{SST}_{AD} are the constant, linear, and quadratic loss coefficients of the AC/DC stage, respectively.

Since the proposed planning model focuses on active-power transfer, the converter current of the AC/DC stage is equivalently represented by:

$$I^{SST}_{AD,t} = \frac{P^{SST}_{t,b}}{\sqrt{3} V_b} \tag{5}$$

where V_b is the rated line-to-line voltage at the 10 kV/AC side.

Similarly, the loss of the DC/DC stage is expressed as:

$$P^{SST}_{loss,DD,t} = \alpha^{SST}_{DD} + \beta^{SST}_{DD} I^{SST}_{DD,t} + \gamma^{SST}_{DD} \left(I^{SST}_{DD,t}\right)^2 \tag{6}$$

where α^{SST}_{DD}, β^{SST}_{DD} and γ^{SST}_{DD} are the loss coefficients of the DC/DC stage.

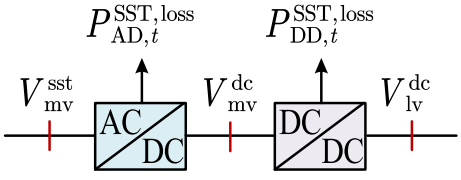


Fig. 3. Loss locations in the SST power conversion stages.

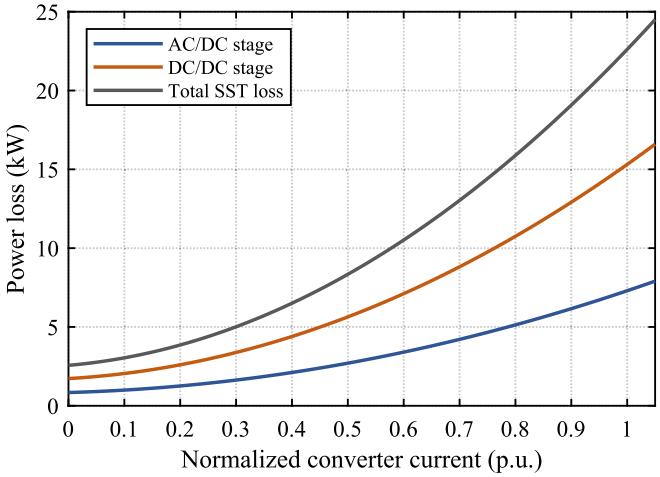


Fig. 4. Representative loss characteristics of the AC/DC and DC/DC conversion stages under different converter loading levels.

These coefficients are determined in the same way as those of the AC/DC stage.

The equivalent current of the DC/DC stage is calculated by:

$$I^{\mathrm{SST}}_{\mathrm{DD},t} = \lambda_{pri}\frac{P^{\mathrm{SST}}_{\mathrm{DD},t}}{V^{\mathrm{dc}}_{\mathrm{mv}}} + \lambda_{sec}\frac{P^{\mathrm{SST}}_{t,c}}{V^{\mathrm{dc}}_{\mathrm{lv}}} \tag{7}$$

where λ_{pri} and λ_{sec} are current referral coefficients. $V^{\mathrm{dc}}_{\mathrm{mv}}$ is the rated medium-voltage dc-link voltage and $V^{\mathrm{dc}}_{\mathrm{lv}}$ is the rated low-voltage dc-link voltage. The specific MVDC voltage can be specified according to the adopted SST implementation.

Accordingly, the total SST conversion loss at time interval t can be written as:

$$P^{\mathrm{SST,loss}}_{t} = P^{\mathrm{SST,loss}}_{\mathrm{AD},t} + P^{\mathrm{SST,loss}}_{\mathrm{DD},t} \tag{8}$$

Finally, the transferred powers of the two conversion stages are constrained by their installed capacities. The capacity limits of the AC/DC and DC/DC stages are expressed as:

$$\left(P^{\mathrm{SST}}_{t,b}\right)^2 + \left(Q^{\mathrm{SST}}_{t,b}\right)^2 \leqslant \left(S^{\mathrm{SST}}_{\mathrm{AD}}\right)^2 \tag{9}$$

$$|P^{\mathrm{SST}}_{\mathrm{DD},t}| \leqslant S^{\mathrm{SST}}_{\mathrm{DD}} \tag{10}$$

where $Q^{\mathrm{SST}}_{t,b}$ denotes the reactive power exchanged between the SST and the utility-side AC system. $S^{\mathrm{SST}}_{\mathrm{AD}}$ and $S^{\mathrm{SST}}_{\mathrm{DD}}$ denote the installed capacities of the AC/DC stage and the DC/DC stage, respectively.

In this study, the loss coefficients of the AC/DC and DC/DC stages are obtained from the representative SST loss profile reported in [44] and are used to capture the planning-level effects of load-dependent conversion losses. The selection of these coefficients may affect the calculated SST losses, capacity decisions, and resulting planning cost because they depend on converter rating, topology, and device technology. If manufacturer-specific or topology-specific loss data or efficiency curves are available, the coefficients can be refitted accordingly, or the corresponding loss curves can be directly substituted without changing the proposed optimization framework.

2.3. Power consumption characterization of the data center

The power consumption of the data center mainly consists of the power consumption of IT equipment and the strongly coupled auxiliary power consumption associated with cooling and power distribution systems. Let $P^{\mathrm{IT}}_{s,n,t}$ denote the active power consumption of IT equipment at node n during time interval t under scenario s. The aggregated IT power consumption of different server types can be expressed as:

$$P^{\mathrm{IT}}_{s,n,t} = \sum_{k=1}^{K} N^{\mathrm{srv}}_{n,k} P^{\mathrm{srv}}_{s,n,k,t} \tag{11}$$

where k is the server-type index, K is the number of server types, $N^{\mathrm{srv}}_{n,k}$ denotes the number of type-k servers installed at node n, and $P^{\mathrm{srv}}_{s,n,k,t}$ denotes the power consumption of a single type-k server at node n during time interval t under scenario s.

To characterize discrete server deployment while maintaining a compact formulation, the installed number of servers is represented through a binary expansion form:

$$N^{\mathrm{srv}}_{n,k} = \sum_{\alpha=1}^{B_{n,k}} 2^{\alpha-1} z_{n,k,\alpha} N^{\mathrm{mod}}_{n,k} \tag{12}$$

where $N^{\mathrm{mod}}_{n,k}$ denotes the number of type-k servers contained in one unit module at node n, $z_{n,k,\alpha}$ is a binary auxiliary variable indicating whether the α-th module is selected, and $B_{n,k}$ is the number of binary variables used to describe the maximum configurable server scale.

The power consumption of a single server consists of a fixed component and a computing-related component, which is given by:

$$P^{\mathrm{srv}}_{s,n,k,t} = P^{\mathrm{fix}}_{n,k} + P^{\mathrm{cmp}}_{s,n,k,t} \tag{13}$$

where $P^{\mathrm{fix}}_{n,k}$ denotes the fixed power consumption of a type-k server at node n, and $P^{\mathrm{cmp}}_{s,n,k,t}$ denotes the computing-related power consumption associated with the processing units.

Considering the DVFS mechanism, the computing-related power consumption of a server is frequency-dependent and can be modeled as:

$$P^{\mathrm{cmp}}_{s,n,k,t} = \kappa_1 \sum_{l=1}^{L} u_{s,n,k,t,l} \left(f^{\mathrm{cmp}}_{n,k,l}\right)^2 \tag{14}$$

where l is the index of the discrete computing-frequency level, L is the number of discrete frequency levels, $u_{s,n,k,t,l}$ is the frequency selection variable, $f^{\mathrm{cmp}}_{n,k,l}$ denotes the computing frequency corresponding to level l, and κ_1 is power consumption coefficient. To ensure that only one operating frequency is selected at each time interval, the following constraint is imposed:

$$\sum_{l=1}^{L} u_{s,n,k,t,l} = 1, \quad u_{s,n,k,t,l} \in \{0,1\} \tag{15}$$

Accordingly, the actual operating frequency of the computing units can be expressed as:

$$f^{\mathrm{cmp}}_{s,n,k,t} = \sum_{l=1}^{L} u_{s,n,k,t,l} f^{\mathrm{cmp}}_{n,k,l} \tag{16}$$

where $f^{\mathrm{cmp}}_{s,n,k,t}$ denotes the actual operating frequency of the computing units in a type-k server at node n during time interval t under scenario s.

The available computing capacity $H^{\mathrm{cmp}}_{s,n,k,t}$ is determined by the installed server quantity and the service efficiency of each server, which can be expressed as:

$$H^{\mathrm{cmp}}_{s,n,k,t} = \varphi_{s,n,k,t} N^{\mathrm{srv}}_{n,k} \tag{17}$$

$$\varphi_{s,n,k,t} = \kappa_2 f^{\mathrm{cmp}}_{s,n,k,t} \tag{18}$$

where $\varphi_{s,n,k,t}$ denotes the service efficiency of type-k servers at node n during time interval t under scenario s, and κ_2 is a proportional coefficient that maps the operating frequency to computing service capacity.

Furthermore, the power usage effectiveness (PUE) is introduced to characterize the auxiliary power consumption associated with cooling and power distribution systems. Let $P^{\mathrm{tot}}_{s,n,t}$ denote the total power con-

sumption of the computing equipment room at node n, including both IT and auxiliary power consumption. Then, it can be expressed as:

$$P^{\mathrm{tot}}_{s,n,t} = \eta^{\mathrm{PUE}} P^{\mathrm{IT}}_{s,n,t} \tag{19}$$

where η^{PUE} denotes the PUE of the DC data center. Accordingly, the auxiliary power consumption can be written as:

$$P^{\mathrm{aux}}_{s,n,t} = \left(\eta^{\mathrm{PUE}} - 1\right) P^{\mathrm{IT}}_{s,n,t} \tag{20}$$

Therefore, the proposed model captures the coupling relationships among server deployment, DVFS-enabled operation, computing capability, and auxiliary facilities, thereby providing the basis for the subsequent workload flexibility modeling and planning formulation.

2.4. Tiered workload flexibility modeling

To characterize workload flexibility under different operating scenarios, incoming workloads are classified into delay-sensitive and delay-tolerant categories according to their latency requirements, and different scheduling constraints are imposed accordingly. The workload-type index ρ is further divided into several classes according to the stringency of the service level objective (SLO) [45,46], thereby representing different levels of latency tolerance, as summarized in Table 2. The listed SLO classes and delay-tolerance parameters are representative settings used to characterize differentiated workload flexibility. Accordingly, the following formulations are established for delay-sensitive and delay-tolerant workloads.

For the workload class with $\rho = 1$, which represents delay-sensitive services, processing must be completed within the same time interval in which the requests arrive. Therefore, the processed amount of type-ρ delay-sensitive workload at node n during time interval t under scenario s, denoted by $p^{\mathrm{cmp}}_{s,n,\rho,t}$, is equal to the total amount transmitted from all front-end servers:

$$p^{\mathrm{cmp}}_{s,n,\rho,t} = \sum_{m=1}^{M} a^{\mathrm{cmp}}_{s,n,\rho,m,t} \tag{21}$$

where $a^{\mathrm{cmp}}_{s,n,\rho,m,t}$ denotes the amount of type-ρ delay-sensitive workload transmitted from front-end server m to the data center at node n during time interval t under scenario s, and M is the number of front-end servers.

To enforce the latency requirement of delay-sensitive services, an M/M/1 queueing model is adopted as a tractable latency approximation. For type-ρ workload assigned to type-k servers, the average workload assigned to each installed server can be represented by $p^{\mathrm{cmp}}_{s,n,k,\rho,t}/N^{\mathrm{srv}}_{n,k}$. According to the M/M/1 model, the average response time is determined by the difference between the service rate and the arrival rate. Therefore, requiring the response time to be no larger than the maximum tolerable delay $\tau^{\max}_{\rho}$ leads to the following latency margin constraint:

$$v^{\mathrm{srv}}_{s,n,k,t} - \frac{p^{\mathrm{cmp}}_{s,n,k,\rho,t}}{N^{\mathrm{srv}}_{n,k}} \geqslant \frac{1}{\tau^{max}_{\rho}} \tag{22}$$

where k is the server-type index, $p^{\mathrm{cmp}}_{s,n,k,\rho,t}$ denotes the amount of type-ρ workload assigned to type-k servers at node n during time interval t under scenario s. The aggregated processed workload $p^{\mathrm{cmp}}_{s,n,\rho,t}$ is obtained by summing $p^{\mathrm{cmp}}_{s,n,k,\rho,t}$ over all server types. $v^{\mathrm{srv}}_{s,n,k,t}$ is the service rate of type-k servers, and τ^{max}_{ρ} is the maximum tolerable delay of type-ρ workload.

For delay-tolerant workloads, inter-temporal shifting is allowed within the corresponding delay tolerance window. Accordingly, for $\rho = 2,3,4$, the mismatch between the arriving and processed delay-tolerant workloads is represented by the temporal shifting amount $\Delta q^{\mathrm{cmp}}_{s,n,\rho,t}$, which is written as:

$$\Delta q^{\mathrm{cmp}}_{s,n,\rho,t} = \sum_{m=1}^{M} a^{\mathrm{cmp}}_{s,n,\rho,m,t} - p^{\mathrm{cmp}}_{s,n,\rho,t} \tag{23}$$

To describe the accumulation and release of deferred workloads across time intervals, the stored workload, equivalently viewed as the backlog queue, is denoted by $q^{\mathrm{cmp}}_{s,n,\rho,t}$ and modeled as:

$$q^{\mathrm{cmp}}_{s,n,\rho,t} = q^{\mathrm{cmp}}_{s,n,\rho,t-1} + \Delta q^{\mathrm{cmp}}_{s,n,\rho,t} \tag{24}$$

The backlog variable is further constrained by:

$$q^{\mathrm{cmp}}_{s,n,\rho,T} = q^{\mathrm{cmp}}_{s,n,\rho,t_0} = 0, \quad q^{\mathrm{cmp}}_{s,n,\rho,t} \geqslant 0 \tag{25}$$

$$0 \leqslant \sum_{\rho=1}^{Q} q^{\mathrm{cmp}}_{s,n,\rho,t} \leqslant q^{\max}_{n} \tag{26}$$

where t_0 and T denote the beginning and the end of the scheduling horizon, respectively, Q is the number of workload types, and $q^{\max}_{n}$ is the maximum allowable stored workload at node n. The terminal equality constraint prevents artificial workload accumulation or depletion over the scheduling horizon.

In addition, to guarantee that each type of delay-tolerant workload is completed within its tolerable delay window, a cumulative processing constraint is imposed at each deadline instant $\hat{t}$:

$$\sum_{t=t_0}^{\hat{t}} \sum_{n=1}^{N} p^{\mathrm{cmp}}_{s,n,\rho,t} \geqslant A^{\mathrm{req}}_{s,\rho,\hat{t}} \tag{27}$$

where N is the number of nodes and $A^{\mathrm{req}}_{s,\rho,\hat{t}}$ denotes the required cumulative amount of type-ρ delay-tolerant workload that must be completed by time interval $\hat{t}$. This required cumulative amount is determined by the tolerable delay duration $\tau^{\mathrm{tol}}_{\rho}$ as:

$$A^{\mathrm{req}}_{s,\rho,\hat{t}} = \begin{cases} \sum_{t=t_0}^{\hat{t}-\tau^{\mathrm{tol}}_{\rho}} \sum_{n=1}^{N} \sum_{m=1}^{M} a^{\mathrm{cmp}}_{s,n,\rho,m,t} & \hat{t} - t_0 \geqslant \tau^{\mathrm{tol}}_{\rho} \\ 0 & \hat{t} - t_0 < \tau^{\mathrm{tol}}_{\rho} \end{cases} \tag{28}$$

Table 2
SLO-based workload classification and delay tolerance parameters.

Workload type ρ	SLO level	Workload category	Typical service type	Typical flexibility mode	Suggested delay-tolerance parameter
1	P99	Latency-sensitive	Online inference, interactive services, real-time requests	No temporal shifting; must be served in the current slot	$\tau_1 = 0$
2	P95	Slightly delay-tolerant	Soft real-time analytics, non-critical inference tasks	Short-term DVFS or mild throttling	$\tau_2 = 1 \sim 2\Delta t$
3	P90	Moderately delay-tolerant	Batch subtasks, checkpoint-restart jobs, migratable training tasks	Pause-resume or workload migration	$\tau_3 = 2 \sim 4\Delta t$
4	P80	Highly delay-tolerant	Offline training, backfill jobs, batch data processing	Deferred execution or spatiotemporal shifting	$\tau_4 = 4 \sim 12\Delta t$

where $\tau_{\rho}^{\mathrm{tol}}$ is the tolerable delay duration of type-ρ workload.

To guarantee sufficient computing capability, the total amount of workloads processed by each server type should not exceed the secure operating limit of the available computing resource:

$$\sum_{\rho=1}^{Q} \frac{p_{s,n,k,\rho,t}^{\mathrm{cmp}}}{\omega_{k,\rho}} \leqslant \beta^{\mathrm{sec}} H_{s,n,k,t}^{\mathrm{cmp}} \tag{29}$$

where $\omega_{k,\rho}$ denotes the relative processing capability coefficient of type-k servers for type-ρ workloads, $H_{s,n,k,t}^{\mathrm{cmp}}$ denotes the available computing resource, and β^{sec} is the secure operating factor. A larger $\omega_{k,\rho}$ indicates that less equivalent computing capacity is required to process the same amount of workload. In this paper, $\beta^{\mathrm{sec}} = 0.9$, which represents the reserved capacity margin for secure operation. At the 15 min planning resolution, workload arrivals, processed workloads, deferred workloads, and computing capacities are represented on a consistent normalized computing-workload basis. The processing capability of one server is measured in normalized workload units per server per interval, while the total available computing capacity is measured in normalized workload units per interval.

Through the above formulation, the proposed model captures both the strict real-time requirements of delay-sensitive services and the inter-temporal schedulability of delay-tolerant services. Delay-sensitive workloads must be processed in their arrival intervals, whereas delay-tolerant workloads can be rescheduled only within their corresponding service level delay windows. After the processing time is determined, the workloads are allocated among heterogeneous CPU and GPU servers according to their workload-dependent relative processing capabilities. The resulting equivalent computing-resource requirements are constrained by the secure computing capacity provided by the installed servers.

The available computing capacity is jointly determined by the installed server quantities and their DVFS-dependent operating frequencies, while the selected frequencies also determine the computing-related power consumption of individual servers. By aggregating the power consumption of all installed servers and accounting for auxiliary power consumption through the PUE factor, the scheduled workloads are mapped to the IT power demand and total data center power demand. Accordingly, SLO-constrained workload scheduling, workload assignment among heterogeneous servers, workload-dependent relative processing capabilities, server deployment, and DVFS operation jointly establish the relationship between workload demand and power consumption, thereby coupling computing-side flexibility with power-side resource planning.

2.5. Energy storage system operation

To ensure a high level of power supply reliability for critical data center services, battery energy storage systems are deployed within the data center as fast-response backup and balancing resources. In the proposed model, the operational state of the energy storage system is described by its stored energy, which evolves with the charging and discharging processes over time. Let $E_{s,n,t}^{\mathrm{ess}}$ denote the energy state of the storage system at node n during time interval t under scenario s, $P_{s,n,t}^{\mathrm{ch}}$ and $P_{s,n,t}^{\mathrm{dis}}$ denote the charging and discharging power, respectively. The state transition of the storage system can be expressed as:

$$E_{s,n,t+1}^{\mathrm{ESS}} = E_{s,n,t}^{\mathrm{ESS}} + \left(\eta^{\mathrm{ch}} P_{s,n,t}^{\mathrm{ch}} - P_{s,n,t}^{\mathrm{dis}} \Big/ \eta^{\mathrm{dis}}\right) \Delta t \tag{30}$$

where η^{ch} and η^{dis} are the charging and discharging efficiencies, respectively, and Δt is the time-step length. This formulation captures the dynamic coupling between power exchange and the stored energy level of the battery system.

The energy state is bounded by the installed storage capacity:

$$0 \leqslant E_{s,n,t}^{\mathrm{ESS}} \leqslant E_{n}^{\mathrm{ESS}} \tag{31}$$

To avoid artificial energy depletion or accumulation over the scheduling horizon, the terminal energy state is constrained to be equal to the initial one:

$$E_{s,n,T}^{\mathrm{ESS}} = E_{s,n,t_0}^{\mathrm{ESS}} \tag{32}$$

where t_0 and T denote the beginning and the end of the scheduling horizon, respectively.

In addition, the charging and discharging power are limited by their maximum allowable values:

$$0 \leqslant P_{s,n,t}^{\mathrm{ch}} \leqslant u_{s,n,t}^{\mathrm{ch}} \overline{P}_{n}^{\mathrm{ch}} \tag{33}$$

$$0 \leqslant P_{s,n,t}^{\mathrm{dis}} \leqslant u_{s,n,t}^{\mathrm{dis}} \overline{P}_{n}^{\mathrm{dis}} \tag{34}$$

where $\overline{P}_{n}^{\mathrm{ch}}$ and $\overline{P}_{n}^{\mathrm{dis}}$ are the maximum charging and discharging power limits, which are determined according to the planned ESS capacity.

To prevent simultaneous charging and discharging within the same time interval, the mutually exclusive operating constraint is imposed as:

$$u_{s,n,t}^{\mathrm{ch}} + u_{s,n,t}^{\mathrm{dis}} \leqslant 1 \tag{35}$$

where $u_{s,n,t}^{\mathrm{ch}}$ and $u_{s,n,t}^{\mathrm{dis}}$ denote the charging and discharging status variables of the ESS, respectively; $u_{s,n,t}^{\mathrm{ch}}, u_{s,n,t}^{\mathrm{dis}} \in \{0,1\}$.

Through the above formulation, the storage system can be coordinated with other power resources to provide energy balancing and emergency support while satisfying its physical operating limits.

2.6. Renewable generation modeling

The integration of renewable generation provides an effective means of reducing the operating cost of the data center campus and improving its overall economic performance. In the proposed architecture, photovoltaic and wind generation are incorporated as local renewable resources, whose temporal output characteristics can be coordinated with workload flexibility to enhance renewable energy utilization.

Let $P_{s,n,t}^{\mathrm{PV}}$ and $P_{s,n,t}^{\mathrm{WT}}$ denote the active power output of the photovoltaic and wind generation units at node n during time interval t under scenario s, respectively. Their outputs are constrained by the installed capacities S_{n}^{PV} and S_{n}^{WT}, as well as the corresponding time-varying renewable availability profiles. The output constraints can be expressed as:

$$0 \leqslant P_{s,n,t}^{\mathrm{PV}} \leqslant \overline{p}_{s,n,t}^{\mathrm{PV}} S_{n}^{\mathrm{PV}} \tag{36}$$

$$0 \leqslant P_{s,n,t}^{\mathrm{WT}} \leqslant \overline{p}_{s,n,t}^{\mathrm{WT}} S_{n}^{\mathrm{WT}} \tag{37}$$

where $\overline{p}_{s,n,t}^{\mathrm{PV}}$ and $\overline{p}_{s,n,t}^{\mathrm{WT}}$ denote the scenario-dependent forecasted availability factors of photovoltaic and wind generation, respectively.

2.7. Diesel generator operation

Diesel generators are installed as backup power sources to enhance the supply reliability of the data center under utility outages or power shortage conditions. Since diesel generators can be started within a relatively short time, their start-up time is neglected in the 15 min scheduling resolution adopted in this study. Let $P_{s,n,t}^{\mathrm{DG}}$ denote the active power output of the diesel generator at node n during time interval t under scenario s. The output of the diesel generator is constrained by its minimum and maximum operating limits:

$$\overline{P}_{n}^{\mathrm{DG}} y_{s,n,t}^{\mathrm{DG}} \leqslant P_{s,n,t}^{\mathrm{DG}} \leqslant \overline{P}_{n}^{\mathrm{DG}} y_{s,n,t}^{\mathrm{DG}} \tag{38}$$

where $y_{s,n,t}^{\mathrm{DG}}$ denote its binary operating status. $y_{s,n,t}^{\mathrm{DG}}=1$ indicates that the diesel generator is in operation during time interval t. $\overline{P}_n^{\mathrm{DG}}$ and $\overline{P}_n^{\mathrm{DG}}$ denote its minimum and maximum output limits, respectively.

To avoid efficiency degradation and lifetime reduction caused by frequent start-stop actions, the diesel generator is required to remain online for at least its minimum continuous operating duration once it is started. Let $T_{\min}^{\mathrm{DG}}$ denote the minimum up-time, and let $u_{s,n,\tau}^{DG}$ denote the start-up indicator. The corresponding constraint can be written as:

$$\sum_{\tau=t}^{t+T_{\min}^{\mathrm{DG}}-1} y_{s,n,\tau}^{DG} \geqslant T_{\min}^{\mathrm{DG}} u_{s,n,t}^{\mathrm{DG}} \tag{39}$$

Meanwhile, the start-up and shut-down status variables are linked to the operating status through:

$$u_{s,n,t}^{\mathrm{DG}} \geqslant y_{s,n,t}^{\mathrm{DG}} - y_{s,n,t-1}^{\mathrm{DG}} \tag{40}$$

$$v_{s,n,t}^{\mathrm{DG}} \geqslant y_{s,n,t-1}^{\mathrm{DG}} - y_{s,n,t}^{\mathrm{DG}} \tag{41}$$

where $v_{s,n,t}^{\mathrm{DG}}$ denotes the shut-down indicator. In addition, to account for the output transition limits during consecutive time intervals, the ramping constraints of the diesel generator are given by:

$$P_{s,n,t}^{\mathrm{DG}} - P_{s,n,t-1}^{\mathrm{DG}} \leqslant R_n^{\mathrm{DG,up}} y_{s,n,t-1}^{\mathrm{DG}} + \overline{P}_n^{\mathrm{DG}} u_{s,n,t}^{\mathrm{DG}} \tag{42}$$

$$P_{s,n,t-1}^{\mathrm{DG}} - P_{s,n,t}^{\mathrm{DG}} \leqslant R_n^{\mathrm{DG,down}} y_{s,n,t}^{\mathrm{DG}} + \overline{P}_n^{\mathrm{DG}} v_{s,n,t}^{\mathrm{DG}} \tag{43}$$

where $R_n^{\mathrm{DG,up}}$ and $R_n^{\mathrm{DG,down}}$ denote the ramp-up and ramp-down limits of the diesel generator at node n, respectively.

3. Bi-level coordinated resource planning model for DC data centers

In a DC data center, the utility grid, renewable generation, energy storage, and backup power sources are coordinated through the SST and internal DC distribution network to supply computing workloads efficiently and reliably. Workload flexibility further enhances operational adaptability on the computing side, strengthening the coupling between power and computing resources. Since long-term investment decisions and scenario-dependent operating schedules are inherently interdependent, these interactions should be explicitly captured in the planning formulation. Therefore, this section establishes a bi-level coordinated planning model, where the upper level determines long-term resource configuration and the lower level derives operating strategies under different scenarios. The objective functions and associated constraints are presented as follows.

3.1. Upper-level planning objective

The upper level determines the long-term planning variables of the DC data center, including the installed capacities of the AC/DC and DC/DC stages of the SST, the energy storage capacity, the installed capacities of PV, WT and DG, as well as the deployment quantity of different server types. These planning variables can be written as:

$$\boldsymbol{x} = \left\{ S_{\mathrm{AD}}^{\mathrm{SST}}, S_{\mathrm{DD}}^{\mathrm{SST}}, E_n^{\mathrm{ESS}}, S_n^{\mathrm{PV}}, S_n^{\mathrm{WT}}, S_n^{\mathrm{DG}}, N_{n,k}^{\mathrm{srv}} \right\} \tag{44}$$

The upper-level objective is formulated as the minimization of the annualized capital cost, which can be written as:

$$\min f_1 = C_{\mathrm{ALC}} \tag{45}$$

where C_{ALC} denotes the annualized capital cost of resource installation.

The annualized capital cost is decomposed into the annualized investment costs of different types of equipment:

$$C_{\mathrm{ALC}} = C_{\mathrm{ALC}}^{\mathrm{SST}} + C_{\mathrm{ALC}}^{\mathrm{ESS}} + C_{\mathrm{ALC}}^{\mathrm{PV}} + C_{\mathrm{ALC}}^{\mathrm{WT}} + C_{\mathrm{ALC}}^{\mathrm{DG}} + C_{\mathrm{ALC}}^{\mathrm{srv}} \tag{46}$$

where $C_{\mathrm{ALC}}^{\mathrm{SST}}$, $C_{\mathrm{ALC}}^{\mathrm{ESS}}$, $C_{\mathrm{ALC}}^{\mathrm{PV}}$, $C_{\mathrm{ALC}}^{\mathrm{WT}}$, $C_{\mathrm{ALC}}^{\mathrm{DG}}$ and $C_{\mathrm{ALC}}^{\mathrm{srv}}$ denote the annualized investment costs of the SST, ESS, PV, WT, DG and server deployment, respectively.

The annualized investment cost of the SST is given by:

$$C_{\mathrm{ALC}}^{\mathrm{SST}} = \frac{\delta(1+\delta)^{\ell}}{(1+\delta)^{\ell}-1} c_{\mathrm{ALC}}^{\mathrm{SST}} \left(S_{\mathrm{AD}}^{\mathrm{SST}} + S_{\mathrm{DD}}^{\mathrm{SST}} \right) \tag{47}$$

where δ is the discount rate, and ℓ is the equipment lifetime. $c_{\mathrm{ALC}}^{\mathrm{SST}}$ is the unit investment cost coefficient of the SST, while $S_{\mathrm{AD}}^{\mathrm{SST}}$ and $S_{\mathrm{DD}}^{\mathrm{SST}}$ denote the planned capacities of the AC/DC and DC/DC converters in the SST, respectively.

Similarly, the annualized investment costs of the energy storage system, photovoltaic generation, wind generation, and diesel generators are given by:

$$C_{\mathrm{ALC}}^{\mathrm{ESS}} = \sum_{n=1}^{N} \frac{\delta(1+\delta)^{\ell}}{(1+\delta)^{\ell}-1} c_{\mathrm{ALC}}^{\mathrm{ESS}} S_n^{\mathrm{ESS}} \tag{48}$$

$$C_{\mathrm{ALC}}^{\mathrm{PV}} = \sum_{n=1}^{N} \frac{\delta(1+\delta)^{\ell}}{(1+\delta)^{\ell}-1} c_{\mathrm{ALC}}^{\mathrm{PV}} S_n^{\mathrm{PV}} \tag{49}$$

$$C_{\mathrm{ALC}}^{\mathrm{WT}} = \sum_{n=1}^{N} \frac{\delta(1+\delta)^{\ell}}{(1+\delta)^{\ell}-1} c_{\mathrm{ALC}}^{\mathrm{WT}} S_n^{\mathrm{WT}} \tag{50}$$

$$C_{\mathrm{ALC}}^{\mathrm{DG}} = \sum_{n=1}^{N} \frac{\delta(1+\delta)^{\ell}}{(1+\delta)^{\ell}-1} c_{\mathrm{ALC}}^{\mathrm{DG}} S_n^{\mathrm{DG}} \tag{51}$$

where S_n^{ESS}, S_n^{PV}, S_n^{WT}, and S_n^{DG} denote the installed capacities of the energy storage system, photovoltaic generation, wind generation, and diesel generator at node n, respectively; and $c_{\mathrm{ALC}}^{\mathrm{ESS}}$, $c_{\mathrm{ALC}}^{\mathrm{PV}}$, $c_{\mathrm{ALC}}^{\mathrm{WT}}$, and $c_{\mathrm{ALC}}^{\mathrm{DG}}$ are the corresponding unit investment cost coefficients.

In addition to power supply resources, the deployment cost of heterogeneous servers is also considered in the upper-level planning objective. Since CPU and GPU servers have different hardware costs and computing capabilities, the unit investment cost is distinguished by server type. The annualized investment cost of server deployment is expressed as:

$$C_{\mathrm{ALC}}^{\mathrm{srv}} = \sum_{n=1}^{N} \sum_{k=1}^{K} \frac{\delta(1+\delta)^{l}}{(1+\delta)^{l}-1} c_{\mathrm{ALC},k}^{\mathrm{srv}} N_{n,k}^{\mathrm{srv}} \tag{52}$$

where $c_{\mathrm{ALC},k}^{\mathrm{srv}}$ denotes the unit investment cost coefficient of type-k servers.

3.2. Lower-level operational objective

Given the planning decisions obtained from the upper level, the lower level optimizes the operating schedules of the DC data center under representative operating scenarios. The lower-level decision variables include the outputs of the SST, ESS, PV, WT, DG, and utility power purchase, as well as the workload processing and scheduling variables associated with the data center. These optimization variables can be written as:

$$\boldsymbol{y}_s = \left\{ \begin{array}{c} P_{s,n,\rho,t}^{\mathrm{ds}}, P_{s,n,\rho,t}^{\mathrm{dt}}, P_{s,n,k,\rho,t}^{\mathrm{ds/dt}}, \Delta q_{s,n,\rho,t}^{\mathrm{dt}}, q_{s,n,\rho,t}^{\mathrm{dt}}, \\ E_{s,n,t}^{\mathrm{ESS}}, P_{s,n,t}^{\mathrm{ch}}, P_{s,n,t}^{\mathrm{dis}}, P_{s,n,t}^{\mathrm{PV}}, P_{s,n,t}^{\mathrm{WT}}, P_{s,n,t}^{\mathrm{DG}}, P_{s,n,t}^{\mathrm{UG}}, \\ P_{s,n,t}^{\mathrm{IT}}, P_{s,n,t}^{\mathrm{aux}}, \Delta P_{s,n,t}^{\mathrm{shed}}, P_{s,ij,t}, V_{s,n,t}, P_{s,n,t}, \\ P_{s,n,k,t}^{\mathrm{srv}}, P_{s,n,k,t}^{\mathrm{cmp}}, f_{s,n,k,t}^{\mathrm{cmp}}, H_{s,n,k,t}^{\mathrm{cmp}}, v_{s,n,k,t}^{\mathrm{srv}}, \\ P_{s,t,b}^{\mathrm{SST}}, P_{\mathrm{AD},s,t}^{\mathrm{SST}}, P_{s,t,a}^{\mathrm{SST}}, P_{s,t,c}^{\mathrm{SST}}, P_{\mathrm{DD},s,t}^{\mathrm{SST}} \end{array} \right\} \tag{53}$$

The lower-level objective is formulated as the minimization of the operating cost and the reliability-related cost, which can be written as:

$$\min f_2 = C_{\mathrm{OPE}} + C_{\mathrm{REL}} \tag{54}$$

where C_{OPE} denotes the operating cost under representative operating scenarios, and C_{REL} denotes the reliability-related loss cost caused by insufficient power supply.

The equipment operating cost is further decomposed into the operating costs of different resources:

$$C_{\mathrm{OPE}} = C_{\mathrm{OPE}}^{\mathrm{SST}} + C_{\mathrm{OPE}}^{\mathrm{ESS}} + C_{\mathrm{curt}}^{\mathrm{PV}} + C_{\mathrm{curt}}^{\mathrm{WT}} + C_{\mathrm{OPE}}^{\mathrm{DG}} + C_{\mathrm{OPE}}^{\mathrm{UG}} \tag{55}$$

Among them, the SST operation and maintenance cost is calculated according to the power transferred through the conversion stages:

$$C_{\mathrm{OPE}}^{\mathrm{SST}} = \sum_{s=1}^{S}\sum_{t=1}^{T} c_{\mathrm{OPE}}^{\mathrm{SST}}\left(P_{\mathrm{AD},s,t}^{\mathrm{SST}} + P_{\mathrm{DD},s,t}^{\mathrm{SST}}\right)\Delta t \tag{56}$$

where $c_{\mathrm{OPE}}^{\mathrm{SST}}$ is the unit operating cost coefficient of the SST.

The operating cost of ESSs is expressed as:

$$C_{\mathrm{OPE}}^{\mathrm{ESS}} = \sum_{s=1}^{S}\sum_{t=1}^{T}\sum_{n=1}^{N} c_t^{\mathrm{ESS}}\left(P_{s,n,t}^{\mathrm{ch}} - P_{s,n,t}^{\mathrm{dis}}\right)\Delta t \tag{57}$$

where c_t^{ESS} is the storage operating cost coefficient at time interval t.

The photovoltaic and wind curtailment costs are calculated as:

$$C_{\mathrm{curt}}^{\mathrm{PV}} = \sum_{s=1}^{S}\sum_{t=1}^{T}\sum_{n=1}^{N} c_{\mathrm{curt}}^{\mathrm{PV}}\left(\overline{p}_{s,n,t}^{\mathrm{PV}} S_n^{\mathrm{PV}} - P_{s,n,t}^{\mathrm{PV}}\right)\Delta t \tag{58}$$

$$C_{\mathrm{curt}}^{\mathrm{WT}} = \sum_{s=1}^{S}\sum_{t=1}^{T}\sum_{n=1}^{N} c_{\mathrm{curt}}^{\mathrm{WT}}\left(\overline{p}_{s,n,t}^{\mathrm{WT}} S_n^{\mathrm{WT}} - P_{s,n,t}^{\mathrm{WT}}\right)\Delta t \tag{59}$$

where $c_{\mathrm{curt}}^{\mathrm{PV}}$ and $c_{\mathrm{curt}}^{\mathrm{WT}}$ are the curtailment penalty coefficients of photovoltaic and wind generation, respectively.

The diesel generator operating cost and utility electricity purchasing cost are given by:

$$C_{\mathrm{OPE}}^{\mathrm{DG}} = \sum_{s=1}^{S}\sum_{t=1}^{T}\sum_{n=1}^{N} c_t^{\mathrm{DG}} P_{s,n,t}^{\mathrm{DG}}\Delta t \tag{60}$$

$$C_{\mathrm{OPE}}^{\mathrm{UG}} = \sum_{s=1}^{N_s}\sum_{t=1}^{N_t}\sum_{n=1}^{N_n} c_t^{\mathrm{UG}} P_{s,n,t}^{\mathrm{UG}}\Delta t \tag{61}$$

where c_t^{DG} is the unit operating cost coefficient of the diesel generator at time interval t, c_t^{UG} is the unit electricity purchasing cost coefficient, and $P_{s,n,t}^{\mathrm{UG}}$ is the purchased utility power at node n during time interval t under scenario s.

The reliability-related cost is defined as the accumulated penalty caused by insufficient power supply over all scenarios, time intervals, and nodes:

$$C_{\mathrm{REL}} = \sum_{s=1}^{S}\sum_{t=1}^{T}\sum_{n=1}^{N} c^{L}\cdot\Delta P_{s,n,t}^{\mathrm{shed}}\Delta t \tag{62}$$

where c^L is the loss cost coefficient per unit equivalent curtailed load, and $\Delta P_{s,n,t}^{\mathrm{shed}}$ denotes the load shedding at node n during time interval t under scenario s.

3.3. Planning constraints

In the proposed multi-resource planning model for the DC data center, the planning variables include the capacities of the SST conversion stages, the energy storage system, photovoltaic generation, wind generation, and diesel generators, as well as the deployment quantity of heterogeneous servers. The server deployment constraint has been introduced in Section 2.3 through the binary module representation. Therefore, this subsection focuses on the capacity bounds and modular configuration constraints of the power supply resources.

Specifically, the planning capacities of the AC/DC and DC/DC conversion stages in the SST are first constrained to be non-negative:

$$0 \leqslant S_{\mathrm{AD}}^{\mathrm{SST}},\ \ 0 \leqslant S_{\mathrm{DD}}^{\mathrm{SST}} \tag{63}$$

To characterize the discrete configuration feature of SST capacity planning, the capacities of the two conversion stages are further represented as integer multiples of the corresponding unit module capacities:

$$S_{\mathrm{AD}}^{\mathrm{SST}} = k_{\mathrm{AD}}^{\mathrm{SST}} S_{\mathrm{AD,mod}}^{\mathrm{SST}},\ \ S_{\mathrm{DD}}^{\mathrm{SST}} = k_{\mathrm{DD}}^{\mathrm{SST}} S_{\mathrm{DD,mod}}^{\mathrm{SST}} \tag{64}$$

where $S_{\mathrm{AD,mod}}^{\mathrm{SST}}$ and $S_{\mathrm{DD,mod}}^{\mathrm{SST}}$ denote the unit module capacities of the AC/DC and DC/DC conversion stages, respectively. $k_{\mathrm{AD}}^{\mathrm{SST}}$ and $k_{\mathrm{DD}}^{\mathrm{SST}}$ are the integer decision variables representing the numbers of installed modules.

Similarly, the planning capacities of the energy storage system, photovoltaic generation, wind generation, and diesel generators are all restricted by their capacity bounds and modular installation constraints. For the energy storage system, the planning capacity at node n is constrained by:

$$0 \leqslant E_n^{\mathrm{ESS}} \leqslant \overline{E}_n^{\mathrm{ESS}} \tag{65}$$

$$E_n^{\mathrm{ESS}} = k_n^{\mathrm{ESS}} E_{\mathrm{mod}}^{\mathrm{ESS}} \tag{66}$$

For photovoltaic and wind generation, the corresponding constraints are given by:

$$0 \leqslant S_n^{\mathrm{PV}} \leqslant \overline{S}_n^{\mathrm{PV}} \tag{67}$$

$$0 \leqslant S_n^{\mathrm{WT}} \leqslant \overline{S}_n^{\mathrm{WT}} \tag{68}$$

$$S_n^{\mathrm{PV}} = k_n^{\mathrm{PV}} S_{\mathrm{mod}}^{\mathrm{PV}} \tag{69}$$

$$S_n^{\mathrm{WT}} = k_n^{\mathrm{WT}} S_{\mathrm{mod}}^{\mathrm{WT}} \tag{70}$$

Likewise, the diesel generator capacity satisfies:

$$0 \leqslant S_n^{\mathrm{DG}} \leqslant \overline{S}_n^{\mathrm{DG}} \tag{71}$$

$$S_n^{\mathrm{DG}} = k_n^{\mathrm{DG}} S_{\mathrm{mod}}^{\mathrm{DG}} \tag{72}$$

where $\overline{E}_n^{\mathrm{ESS}}$, $\overline{S}_n^{\mathrm{PV}}$, $\overline{S}_n^{\mathrm{WT}}$, and $\overline{S}_n^{\mathrm{DG}}$ denote their corresponding upper bounds; $E_{\mathrm{mod}}^{\mathrm{ESS}}$, $S_{\mathrm{mod}}^{\mathrm{PV}}$, $S_{\mathrm{mod}}^{\mathrm{WT}}$, and $S_{\mathrm{mod}}^{\mathrm{DG}}$ denote the unit module capacities; and k_n^{ESS}, k_n^{PV}, k_n^{WT}, and k_n^{DG} are the corresponding integer decision variables. Through the above formulation, the planning model ensures that the installed capacities of all resources remain within feasible ranges while preserving the modular deployment characteristic required in practical engineering implementation.

3.4. Operational constraints

To characterize the power flow distribution in the DC distribution network of the data center campus, a directed-branch power flow model is adopted. For each node n, the nodal power balance is described by the conservation of branch power flows and the net power injection at that node. Accordingly, the power balance constraint can be written as:

$$\sum_{ij\in L_{\mathrm{dc}}} P_{s,ij,t} + P_{s,j,t} = \sum_{jk\in L_{\mathrm{dc}}} P_{s,jk,t} \tag{73}$$

where L_{dc} denotes the set of DC distribution branches, $P_{s,ij,t}$ denotes the directed power flow of branch ij during time interval t under scenario s, and $P_{s,j,t}$ denotes the net power injection at node j. The term $\sum_{ij\in L_{\mathrm{dc}}} P_{s,ij,t}$ represents the summation over all branches flowing into node j, while

$\sum_{jk\in L_{dc}} P_{s,jk,t}$ represents the summation over all branches flowing out of node j.

In addition, the power flow of each DC distribution branch is limited by its transmission capacity:

$$-\overline{P}_{ij} \leqslant P_{s,ij,t} \leqslant \overline{P}_{ij} \tag{74}$$

where $\overline{P}_{ij}$ denotes the maximum allowable active power flow of branch ij.

The linearized voltage drop relationship between branch power flow and nodal voltage is given by:

$$v_{s,i,t} - v_{s,j,t} = 2r_{ij}P_{s,ij,t}, \quad i,j \in \Omega_{dc}, ij \in L_{dc} \tag{75}$$

where $v_{s,i,t}$ and $v_{s,j,t}$ denote the DC voltage magnitudes of nodes i and j, respectively, and r_{ij} is the equivalent resistance of branch ij. In addition, the nodal voltage is constrained within the allowable operating range:

$$\overline{V}_n \leqslant V_{s,n,t} \leqslant \overline{V}_n, \quad \forall s,n,t \tag{76}$$

where $\overline{V}_n$ and $\overline{V}_n$ denote the lower and upper allowable voltage limits of node i, respectively.

The nodal net power injection is determined by the power outputs of different sources and the load consumption at the corresponding node, which can be expressed as:

$$P_{s,n,t} = P^{UG}_{s,n,t} + P^{DG}_{s,n,t} + P^{PV}_{s,n,t} + P^{WT}_{s,n,t} + P^{ESS}_{s,n,t} - P^{IT}_{s,n,t} - P^{aux}_{s,n,t} + \Delta P^{shed}_{s,n,t}, \quad \forall s,n,t \tag{77}$$

where $P^{UG}_{s,n,t}$ denotes the power purchased from the utility grid and injected into the DC network through SSTs, corresponding to the AC/DC stage input power $P^{SST}_{s,n,t,b}$. $P^{DG}_{s,n,t}$ denotes the diesel generator output, $P^{PV}_{s,n,t}$ and $P^{WT}_{s,n,t}$ denote the photovoltaic and wind power outputs, respectively, and $P^{ESS}_{s,n,t}$ denotes the net power exchanged between the energy storage system and the DC network, which is calculated as $P^{ESS}_{s,n,t} = P^{ESS,dis}_{s,n,t} - P^{ESS,ch}_{s,n,t}$. Moreover, $P^{IT}_{s,n,t}$ denotes the IT power consumption of the data center, and $P^{aux}_{s,n,t}$ denotes the power consumption of auxiliary loads in the campus.

To represent utility outage conditions, a utility availability parameter $\delta^{UG}_{s,t}$ is introduced. It equals 1 when the utility supply is available and equals 0 during a utility outage. Accordingly, the utility power injected into the DC network is constrained by:

$$0 \leqslant P^{UG}_{s,n,t} \leqslant \delta^{UG}_{s,t}\overline{P}^{UG}_{n} \tag{78}$$

Through the above formulation, the DC distribution network model establishes the electrical coupling between source-side injections, load-side demands, and network operating states. In this branch-flow formulation, the numbered topology in Fig. 1 is mapped to the node set Ω_{dc} and branch set L_{dc} according to their modeling roles. Specifically, Ω_{dc} contains the DC-side buses of the internal distribution network, while L_{dc} contains the ordinary DC distribution feeders between these buses. The AC-side interface nodes and SST conversion links shown in Fig. 1 are not included in L_{dc}. Instead, the SST conversion links are modeled separately by the power balance, loss, and capacity constraints introduced in Section 2.2. Therefore, the branch-flow constraints in this subsection are applied only to the internal DC distribution feeders.

4. Decomposition-based solution methodology

To solve the proposed annual bi-level coordinated planning model efficiently, an integrated solution methodology is developed in this section. The overall procedure consists of two main steps. First, the nonlinear SST loss expressions are reformulated into a tractable linear form through piecewise linear approximation, so that the original model can be embedded into a mixed-integer linear optimization framework. Second, by exploiting the natural separation between long-term planning decisions and scenario-dependent operational decisions, an enhanced Benders decomposition method [47] is adopted to improve the computational tractability of the large-scale planning problem. In this way, the proposed method retains the nonlinear SST loss characteristics that vary with converter loading, while enabling efficient solution of the bi-level model under multiple operating scenarios.

4.1. Linear reformulation of SST losses

The losses of the AC/DC and DC/DC conversion stages in the SST are modeled as nonlinear functions of the equivalent converter current. Although such formulations can reflect the operating-point-dependent loss characteristics of power electronic converters, they also introduce nonlinear terms into the coordinated planning model. When these nonlinear loss expressions are further coupled with integer capacity decisions, diesel-generator commitment variables, and workload scheduling variables across multiple scenarios and time periods, the resulting optimization problem becomes difficult to solve directly. Therefore, the SST loss model is first reformulated into a piecewise linear form.

Specifically, for each conversion stage, the feasible range of the equivalent converter current is divided into several breakpoints, and the loss value at each breakpoint is obtained from the adopted loss-current relationship. In the base case, this relationship is given by the quadratic loss function. When manufacturer-specific loss curves are available, the breakpoint loss values can also be generated directly from measured or fitted loss-current data. Therefore, the piecewise linear reformulation is compatible with both literature-based loss coefficients and manufacturer-specific loss data. The actual converter current and the associated power loss are then represented by convex combinations of adjacent breakpoint values. In this way, the nonlinear loss function can be approximated by a set of linear expressions over piecewise segments.

Accordingly, for each SST conversion stage g, the feasible range of equivalent converter current $I^{g}_{i,n,t}$ is divided into several breakpoints $0 = \overline{I}^{g}_{1} < \overline{I}^{g}_{2} < \cdots < \overline{I}^{g}_{R}$, and the corresponding loss value at each breakpoint is precomputed as:

$$\overline{P}^{loss,g}_{s,n,t} = \alpha^{g} + \beta^{g}\overline{I}^{g}_{s,n,t} + \gamma^{g}\left(\overline{I}^{g}_{s,n,t}\right)^{2} \tag{79}$$

Then, the piecewise linear approximation is expressed as:

$$I^{g}_{s,n,t} = \sum_{r=1}^{R} \lambda^{g}_{s,n,t,r}\overline{I}^{g}_{r} \tag{80}$$

$$P^{loss,g}_{s,n,t} = \sum_{r=1}^{R} \lambda^{g}_{s,n,t,r}\overline{P}^{loss,g}_{r} \tag{81}$$

$$\sum_{r=1}^{R} \lambda^{g}_{s,n,t,r} = 1, \quad \lambda^{g}_{s,n,t,r} \geq 0 \tag{82}$$

where $\overline{I}^{g}_{r}$ and $\overline{P}^{loss,g}_{r}$ are the equivalent converter current and the corresponding loss value at the r-th breakpoint, respectively. $\lambda^{g}_{s,n,t,r}$ are the piecewise linearization variables subject to special ordered set of type 2 (SOS2) adjacency conditions. The SOS2 condition ensures that at most two adjacent breakpoint variables can take positive values, thereby preserving the piecewise linear interpolation structure.

Through the above treatment, the nonlinear SST loss expressions are converted into a tractable linearized form, while the load-dependent loss characteristics are still preserved with controllable approximation accuracy. After this reformulation, the SST power balance equations, capacity constraints, and loss terms can be embedded into the planning model within a mixed-integer linear programming framework.

4.2. Compact reformulation and master problem

After the piecewise linear reformulation of the SST losses, the proposed annual planning model can be recast as a large-scale mixed-integer two-stage optimization problem. Since the planning variables are shared by all scenarios, whereas the operating variables are scenario-specific, the problem exhibits a natural decomposable structure. Therefore, an enhanced Benders decomposition framework is adopted to improve the computational tractability of the proposed annual planning model. The planning model can be compactly written as:

$$\min_{\boldsymbol{x},\{\boldsymbol{y}_s,\boldsymbol{z}_s\}} \boldsymbol{c}^\top \boldsymbol{x} + \sum_{s\in S} p_s \left(\boldsymbol{d}_s^\top \boldsymbol{y}_s + \boldsymbol{g}_s^\top \boldsymbol{z}_s\right) \tag{83}$$

s.t.

$$\begin{cases} \boldsymbol{A}\boldsymbol{x} \leqslant \boldsymbol{b} \\ \boldsymbol{T}_s\boldsymbol{x} + \boldsymbol{W}_s\boldsymbol{y}_s + \boldsymbol{V}_s\boldsymbol{z}_s \leqslant \boldsymbol{h}_s \\ \boldsymbol{z}_s \in \{0,1\}^{K_s} \end{cases} \tag{84}$$

where $\boldsymbol{x}$ is the vector of first-stage planning variables, $\boldsymbol{y}_s$ and $\boldsymbol{z}_s$ are the vectors of continuous and binary second-stage operating variables under scenario s, respectively, p_s is the probability of scenario s. $\boldsymbol{c}$, $\boldsymbol{d}_s$, and $\boldsymbol{g}_s$ are the cost coefficient vectors associated with the first-stage planning cost, the continuous second-stage operating cost, and the binary second-stage operating cost, respectively, $\boldsymbol{A}$ and $\boldsymbol{b}$ define the constraints involving only the first-stage variables, and $\boldsymbol{T}_s$, $\boldsymbol{W}_s$, $\boldsymbol{V}_s$, and $\boldsymbol{h}_s$ are the coefficient matrices and vector associated with the scenario-dependent operating constraints.

Specifically, $\boldsymbol{z}_s$ collects the binary operating variables in scenario s, including the DVFS frequency selection variables, the ESS charging and discharging status variables, and the DG commitment variables. According to the variables defined in the previous sections, it can be further written as:

$$\boldsymbol{z}_s = \left\{ u_{s,n,k,t,l}, u_{s,n,t}^{\mathrm{ch}}, u_{s,n,t}^{\mathrm{dis}}, y_{s,n,t}^{\mathrm{DG}}, u_{s,n,t}^{\mathrm{DG}}, v_{s,n,t}^{\mathrm{DG}} \right\} \tag{85}$$

Based on the above compact form, the master problem at iteration ν is formulated as:

$$\min_{\boldsymbol{x},\theta} \boldsymbol{c}^\top \boldsymbol{x} + \sum_{s\in S} p_s \boldsymbol{\theta}_s \tag{86}$$

s.t.

$$\begin{cases} \boldsymbol{A}\boldsymbol{x} \leqslant \boldsymbol{b} \\ \boldsymbol{\theta}_s \geqslant 0 \\ \text{generated enhanced Benders cut} \end{cases} \tag{87}$$

where $\boldsymbol{\theta}_s$ is an auxiliary variable representing the current approximation of the optimal second-stage recourse cost under scenario s. The master problem contains only the planning variables and the accumulated cutting planes, thus providing a lower approximation of the original integrated model. Once a tentative planning solution is obtained from the master problem, the scenario-wise operating subproblems can be solved independently in parallel.

4.3. Scenario subproblems and enhanced cut generation

Different from standard Benders decomposition, the scenario-wise operating model in this study contains binary recourse variables, including diesel-generator commitment variables, ESS charging/discharging mode indicators, and discrete DVFS frequency selection variables. Therefore, classical Benders cuts cannot be directly derived from the dual form of the original mixed-integer subproblem. To address this issue, the enhanced Benders decomposition is introduced. The main idea is to fix the binary recourse variables at their incumbent values and derive valid cuts from the resulting linear representation, while an additional feasibility restoration mechanism is used to eliminate unacceptable tentative planning solutions.

For a tentative master solution $(\widehat{\boldsymbol{x}}, \widehat{\boldsymbol{\theta}}_s)$, the operational feasibility and cost consistency under scenario s are jointly evaluated through the following unified primal subproblem:

$$\min_{\boldsymbol{y}_s,\boldsymbol{z}_s,\boldsymbol{u}_s,\boldsymbol{\xi}_s,\boldsymbol{\zeta}_s} 1^\top \boldsymbol{u}_s + \boldsymbol{\xi}_s + \boldsymbol{\zeta}_s \tag{88}$$

s.t.

$$\begin{cases} \boldsymbol{d}_s^\top \boldsymbol{y}_s + \boldsymbol{g}_s^\top \boldsymbol{z}_s + \boldsymbol{\xi}_s \geqslant \widehat{\boldsymbol{\theta}}_s \\ -\boldsymbol{d}_s^\top \boldsymbol{y}_s - \boldsymbol{g}_s^\top \boldsymbol{z}_s + \boldsymbol{\zeta}_s \geqslant -\widehat{\boldsymbol{\theta}}_s \\ \boldsymbol{W}_s\boldsymbol{y}_s + \boldsymbol{V}_s\boldsymbol{z}_s \leqslant \boldsymbol{h}_s - \boldsymbol{T}_s\widehat{\boldsymbol{x}} + \boldsymbol{u}_s \\ \boldsymbol{u}_s \geqslant 0,\ \ \boldsymbol{\xi}_s \geqslant 0,\ \ \boldsymbol{\zeta}_s \geqslant 0 \end{cases} \tag{89}$$

where $\boldsymbol{u}_s$ is the slack vector associated with the scenario-dependent operating constraints, and $\boldsymbol{\xi}_s$ and $\boldsymbol{\zeta}_s$ are additional nonnegative slack variables introduced to measure the mismatch between the master-estimated recourse cost and the actual operating cost under scenario s. When all slack variables are zero, the tentative master solution is accepted by scenario s; otherwise, additional cuts are required to refine the master problem. This unified formulation enables the feasibility and cost-consistency checks to be handled within a single scenario subproblem and therefore simplifies the iterative solution process.

To derive valid cuts from the mixed-integer scenario subproblem, an auxiliary binary vector $\widetilde{\boldsymbol{z}}_s$ is introduced and constrained by $\boldsymbol{z}_s = \widetilde{\boldsymbol{z}}_s$, so that the inner optimization problem becomes a linear program after relaxing the integrality of $\boldsymbol{z}_s$. By dualizing the resulting inner linear program, the following unified dual subproblem is obtained:

$$F_{D,s}^* = \max_{(\boldsymbol{\mu}_s,\boldsymbol{\nu}_s,\boldsymbol{\omega}_s)\in\mathscr{C}_s\times\mathscr{O}_s} (\overline{\boldsymbol{e}}_s - \overline{\boldsymbol{B}}_s\widehat{\boldsymbol{x}} - \overline{\boldsymbol{g}}_s\widehat{\boldsymbol{\theta}}_s)^\top \boldsymbol{\mu}_s + 1^\top \boldsymbol{\omega}_s \tag{90}$$

where $\boldsymbol{\mu}_s$, $\boldsymbol{\nu}_s$, and $\boldsymbol{\omega}_s$ are the dual variables associated with the unified scenario subproblem, $\mathscr{C}_s$ and $\mathscr{O}_s$ are the corresponding feasible sets, and $\overline{\boldsymbol{B}}_s$, $\overline{\boldsymbol{e}}_s$, and $\overline{\boldsymbol{g}}_s$ are the aggregated coefficient matrices and vectors associated with the unified representation. If $F_{D,s}^* > 0$, a unified Benders cut is generated as:

$$(\overline{\boldsymbol{e}}_s - \overline{\boldsymbol{B}}_s\boldsymbol{x} - \overline{\boldsymbol{g}}_s\boldsymbol{\theta}_s)^\top \widehat{\boldsymbol{\mu}}_s + 1^\top \widehat{\boldsymbol{\omega}}_s \leqslant 0 \tag{91}$$

where $\widehat{\boldsymbol{\mu}}_s$ and $\widehat{\boldsymbol{\omega}}_s$ are the optimal dual solutions of the unified dual subproblem at the current iteration. The generated cut improves the approximation of the recourse cost and tightens the feasible region of the master problem.

However, the presence of binary recourse variables may make the unified Benders cuts insufficient to exclude some unacceptable master solutions. In such cases, a feasibility restoration subproblem is introduced to measure the deviation of the tentative master solution from the scenario-feasible region. It is formulated as:

$$F_{F,s}^* = \min_{\boldsymbol{x}\boldsymbol{y}_s,\boldsymbol{z}_s} \Delta_s(\boldsymbol{x}, \widehat{\boldsymbol{x}}) \tag{92}$$

s.t.

$$\begin{cases} \boldsymbol{T}_s\boldsymbol{x} + \boldsymbol{W}_s\boldsymbol{y}_s + \boldsymbol{V}_s\boldsymbol{z}_s \leqslant \boldsymbol{h}_s \\ \boldsymbol{z}_s \in \{0,1\}^{K_s} \end{cases} \tag{93}$$

where $\Delta_s(\boldsymbol{x}, \widehat{\boldsymbol{x}})$ is a normalized measure of the deviation between the current tentative solution and the nearest scenario-feasible solution. In this study, it is defined as:

$$\Delta_s(\boldsymbol{x}, \widehat{\boldsymbol{x}}) = \sum_{e\in\mathscr{E}_s} \frac{|\boldsymbol{x}_e - \widehat{\boldsymbol{x}}_e|}{\max\{|\widehat{\boldsymbol{x}}_e|, \tau\}} \tag{94}$$

where $\mathscr{E}_s$ denotes the set of first-stage variables that are coupled with scenario s, and τ is a sufficiently small positive constant introduced to avoid division by zero. Based on the optimal value $F_{F,s}^*$, the following

feasibility restoration cut is added to the master problem:

$$\Delta_s(\boldsymbol{x}, \widehat{\boldsymbol{x}}) \geq F^*_{F,s} \tag{95}$$

This cut explicitly excludes the current invalid tentative solution and further tightens the feasible region of the master problem. Consequently, the enhanced Benders decomposition framework adopted here consists of two types of cuts, namely the unified Benders cuts and the feasibility restoration cuts, which jointly improve the iterative refinement of the master problem.

4.4. Enhanced Benders decomposition iterative procedure

The overall solution process of the proposed enhanced Benders decomposition is illustrated in Fig. 5. The master problem provides a tentative planning solution, and the scenario subproblems evaluate this solution in parallel. According to the evaluation results, unified Benders cuts and, if necessary, feasibility restoration cuts are generated and added to the master problem. This process is repeated until no violated cut is generated and the convergence criterion is satisfied. The main steps are summarized as follows.

1) Initialize the master problem without any additional cutting planes and set the iteration counter $\nu = 1$.
2) Solve the master problem to obtain the tentative planning solution $(\widehat{\boldsymbol{x}}^{\nu}, \widehat{\boldsymbol{\theta}}_s^{\nu})$.
3) For the fixed tentative planning solution, solve all scenario-wise unified subproblems in parallel. For each scenario s, first solve the unified dual subproblem. If $F^*_{D,s} > 0$, generate a unified Benders cut and return it to the master problem.
4) If the unified dual subproblem does not yield a violated cut but the tentative solution is still not acceptable, solve the feasibility restoration subproblem and generate a feasibility restoration cut.
5) Add all newly generated cuts to the master problem and update the lower approximation of the original integrated model.
6) Repeat the above process until no new cut is generated and the convergence criterion is satisfied.

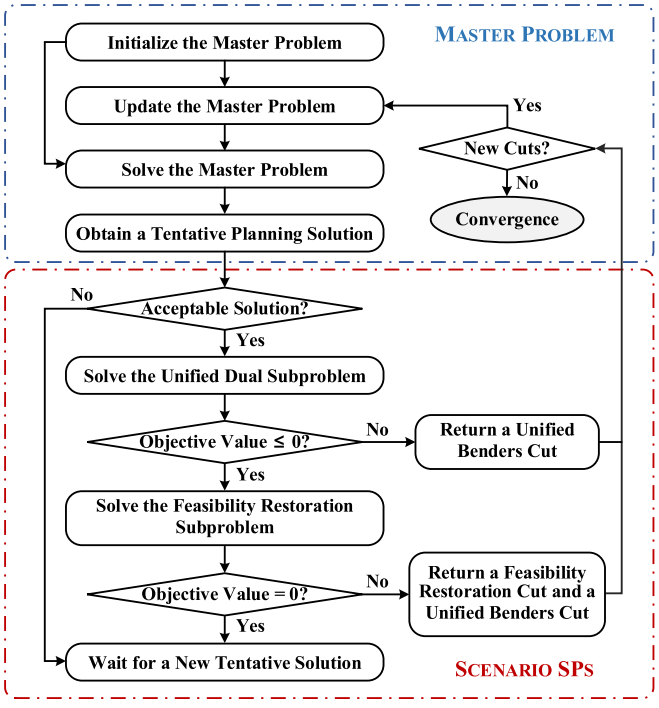


Fig. 5. Flowchart of the proposed enhanced Benders decomposition algorithm.

5. Case study

The optimization model is solved using Gurobi interfaced with the YALMIP toolbox in MATLAB. All simulations are carried out on a Lenovo ThinkCentre desktop computer equipped with an Intel(R) Core(TM) i7–12,700 CPU at 2.10 GHz and 16 GB of RAM.

5.1. System configuration and scenario settings

A modified SST-based DC data center with a 2 N power supply architecture is adopted as the study system. The architecture shown in Fig. 1 is used as a representative implementation for validating the proposed planning method. At the system level, the campus is structured as a DC distribution network interfaced with the upstream utility grid through SSTs. As the core power electronic interfaces, the SSTs establish the electrical coupling between the utility-side AC system and the internal medium-voltage and low-voltage DC networks, thereby constituting the principal energy access and conversion infrastructure of the data center.

Within this representative architecture, photovoltaic generation and wind power are integrated at the medium-voltage DC side, whereas the battery energy storage system is deployed on the load-side DC bus. Through the coordinated interconnection of the SSTs and the internal DC network, utility supply, backup generation, renewable resources, energy storage, and data center loads are incorporated into a unified electrical topology. The workload is divided into delay-sensitive and delay-tolerant types to represent differentiated service requirements and scheduling flexibility. It should be noted that practical SST-based data center architectures may adopt different voltage levels, port configurations, DC bus structures, and resource access positions. The proposed planning framework is not restricted to the specific topology shown in Fig. 1 and can be adapted to other SST-based implementations by updating the network topology, resource access positions, loss characteristics, capacity limits, and cost parameters.

A year-round planning horizon is considered with a time resolution of 15 min, corresponding to 365 daily scenarios and 96 scheduling intervals per day. Normal operating conditions and utility outage scenarios are both included. Utility outage scenarios are constructed by imposing representative outage windows on the chronological profiles, with their duration and starting time determined from historical outage records or representative contingency settings. Under the two-stage planning framework, the first stage determines the installation capacities of SST conversion stages, PV, WT, ESS, diesel generators, and servers, while the second stage optimizes the operation of all resources and workloads under different scenarios.

The data center-related parameters are summarized in Appendix A, while the technical and economic parameters of the power supply equipment are provided in Appendix B. These parameters were selected with reference to existing data center park planning studies and representative data center power supply configuration studies [6,32]. To examine the impacts of utility-supply interruption duration and occurrence frequency on the planning results, the detailed settings of representative outage scenarios are provided in Appendix C.

The CPU and GPU parameters in Appendix A are representative planning-level values used to characterize two heterogeneous server classes. Their frequency levels, fixed power consumption, frequency-dependent power coefficients, relative processing capability coefficients, and unit investment costs are specified separately for the two server types. The parameter settings are informed by existing studies on data center planning and power supply configuration, together with the modeling requirements of the present case study. When model-specific benchmark or measurement data become available, the corresponding processing capability, power, and cost parameters can be further refined or replaced without changing the overall proposed planning framework.

The representative workload, PV output, and WT output profiles used for scenario construction are shown in Fig. 6. Specifically, the

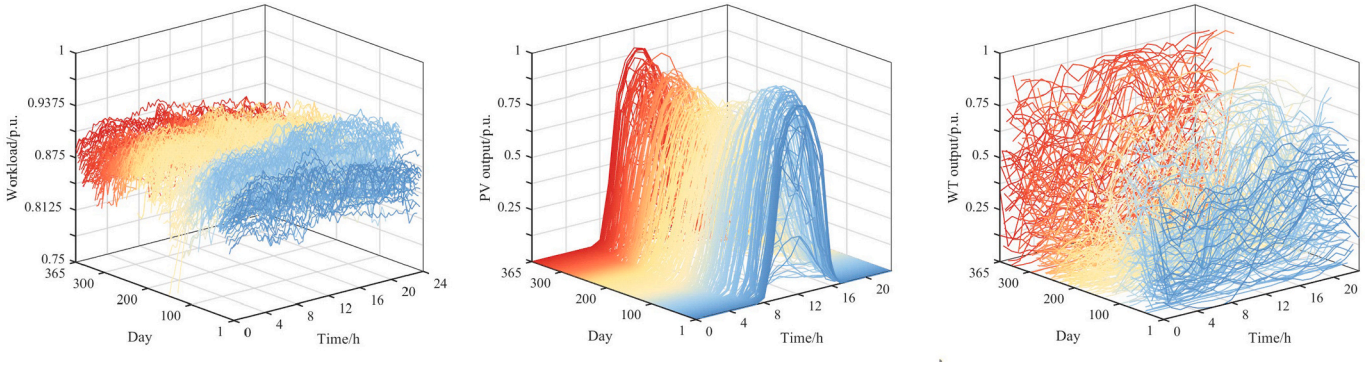


Fig. 6. Representative annual operating profiles. (a) Daily workload profile. (b) Daily PV output profile. (c) Daily WT output profile.

workload profile in Fig. 6(a) was constructed based on an anonymized operational profile of a commercial Internet data center provided by VNET Group, Inc. The profile was used to represent the temporal variation of computing demand in the case study. Since the available operational data describe the aggregated computing demand of the data center rather than fine-grained task-level request traces, the profile was normalized at the 15 min resolution and scaled to a representative request-rate level of 10^5 requests/s for use in the planning model. The PV and WT availability profiles in Fig. 6(b) and Fig. 6(c) were obtained from cleaned historical renewable generation profiles and normalized into per-unit availability factors with respect to their installed capacities.

In addition, a time-of-use electricity tariff is adopted to reflect the temporal variation in electricity purchasing cost, as shown in Fig. 7. The regulation cost of the ESS is 0.34 CNY/kWh. The curtailment penalty costs of PV and WT are both set to 0.95 CNY/kWh.

5.2. Planning results and operational analysis

To evaluate the effectiveness of the proposed coordinated planning model, two schemes are compared in this subsection.

Scheme I: workload flexibility is not considered, and the incoming computing demand is treated as inflexible.

Scheme II: workload flexibility is enabled, and delay-tolerant workloads are allowed to be shifted across time intervals subject to service constraints.

The planning results are summarized in Table 3. Compared with Scheme I, Scheme II reduces the AC/DC and DC/DC capacities from 3.0 MW and 4.0 MW to 2.5 MW and 3.0 MW, respectively. The DG capacity is also reduced from 2.0 MW to 1.5 MW, while the ESS capacity remains unchanged at 2.0 MWh. These results indicate that workload flexibility mainly reduces the capacity requirements associated with peak-period

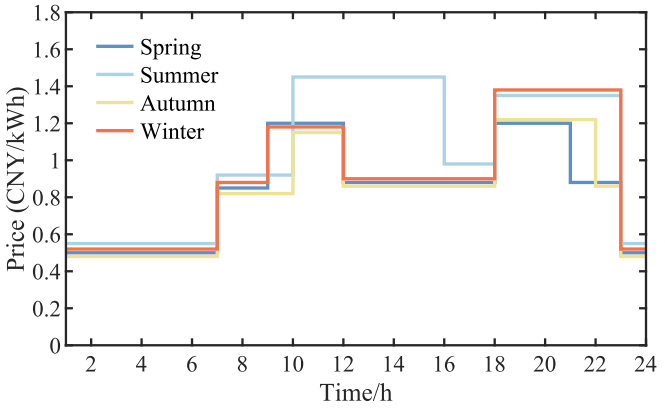


Fig. 7. Representative seasonal time-of-use electricity tariffs.

power conversion and backup supply. From the optimized operating results, the peak total power demand decreases from 3.372 MW in Scheme I to 2.635 MW in Scheme II, corresponding to a reduction of 0.737 MW or 21.85%. Meanwhile, the peak SST DC/DC output decreases from 3.964 MW to 2.973 MW. This is because part of the delay-tolerant workload can be rescheduled from high-demand or supply-constrained intervals to more suitable periods, thereby reducing the instantaneous power requirement that must be supported by SST conversion and backup generation resources. By contrast, the ESS capacity remains unchanged, suggesting that short-term energy buffering is still required for renewable fluctuation smoothing and contingency response, even when computing-side flexibility is available.

The installed PV and WT capacities decrease from 2.0 MW and 6.0 MW in Scheme I to 1.5 MW and 4.5 MW in Scheme II, respectively. This result is related to the change in the overall resource configuration after workload flexibility is introduced. Since flexible workload scheduling reduces peak demand and improves the temporal coordination between computing demand and available supply, the system no longer needs the same level of renewable overcapacity to reduce grid purchases or support reliability requirements. Meanwhile, the lower renewable capacity also helps reduce curtailment costs, as shown in Table 4. Therefore, workload flexibility affects not only backup resource planning, but also the economic sizing of renewable generation.

The server configuration also changes significantly, with the number of CPU servers decreasing from 38,880 to 15,820 and the number of GPU servers increasing from 812 to 3839. In Scheme I, workload processing cannot be shifted across time intervals, and the optimized configuration is dominated by CPU servers under the adopted parameters. In Scheme II, delay-tolerant workloads can be rescheduled within their respective SLO windows and assigned to heterogeneous servers more flexibly. Since GPU servers have a progressively stronger processing-capability advantage for the P95 to P80 workload classes, the model assigns a larger share of these flexible workloads to GPUs. Because the total computing workload remains unchanged, the increase in GPU deployment is accompanied by a substantial reduction in the required number of CPU servers. A more detailed explanation of this heterogeneous server reconfiguration is provided in Section 5.3.

The annual cost breakdown is shown in Table 4. The total annual cost decreases from 68.495 million CNY in Scheme I to 47.808 million CNY in Scheme II, corresponding to a reduction of 30.20%. The largest cost reduction occurs in the annualized investment cost, which includes the annualized capital costs of all planned resources listed in Table 3. This cost decreases from 58.087 million CNY to 39.333 million CNY. This reduction is consistent with the capacity results in Table 3, since smaller SST, DG, renewable, and server configurations directly reduce the annualized capital cost. In addition, the grid electricity cost, backup generation cost, and renewable curtailment cost are all reduced, indicating that workload flexibility also improves the operational coordination among computing demand, electricity purchasing, renewable utilization, and backup generation.

Table 3
Optimal resource planning results.

Scheme	SST capacity (MW)		ESS capacity (MWh)	PV capacity (MW)	WT capacity (MW)	DG capacity (MW)	Number of servers	
	AC/DC	DC/DC	ESS	PV	WT	DG	CPU	GPU
I	3.0	4.0	2.0	2.0	6.0	2.0	38,880	812
II	2.5	3.0	2.0	1.5	4.5	1.5	15,820	3839

Table 4
Annual cost breakdown under different schemes (million CNY).

Scheme	Reliability-related cost	Grid electricity cost	PV curtailment cost	WT curtailment cost	Backup generation cost	Other operating cost	Investment cost	Total cost
I	0.217	8.012	0.385	0.378	0.410	1.007	58.087	68.495
II	0.208	6.523	0.274	0.162	0.280	1.028	39.333	47.808

It is worth noting that the power-related load shedding cost remains very small in both schemes. After workload flexibility is introduced, the annual load shedding cost decreases from 0.217 million CNY in Scheme I to 0.208 million CNY in Scheme II. This indicates that the reduction in planning capacity caused by workload flexibility does not increase the power-side reliability penalty. Instead, flexible workload scheduling helps alleviate instantaneous supply pressure during critical intervals, while the planned SST, ESS, and DG capacities remain sufficient to keep load shedding at a very limited level. Therefore, Scheme II achieves a lower total annual cost mainly through reductions in investment cost, grid electricity cost, renewable curtailment cost, and backup generation cost.

The representative operating profiles of Scheme II are further shown in Fig. 8. Under normal conditions, as shown in Fig. 8(a), the total power consumption is mainly supplied by WT, PV, and the utility grid. PV output is concentrated around midday, while WT provides a relatively stable contribution over the day. The gap between the IT load and the total power consumption reflects the additional power demand caused by auxiliary consumption, ESS charging, and conversion losses. The overall supply-demand balance is maintained without relying heavily on backup generation, indicating that the planned resources can support normal operation economically.

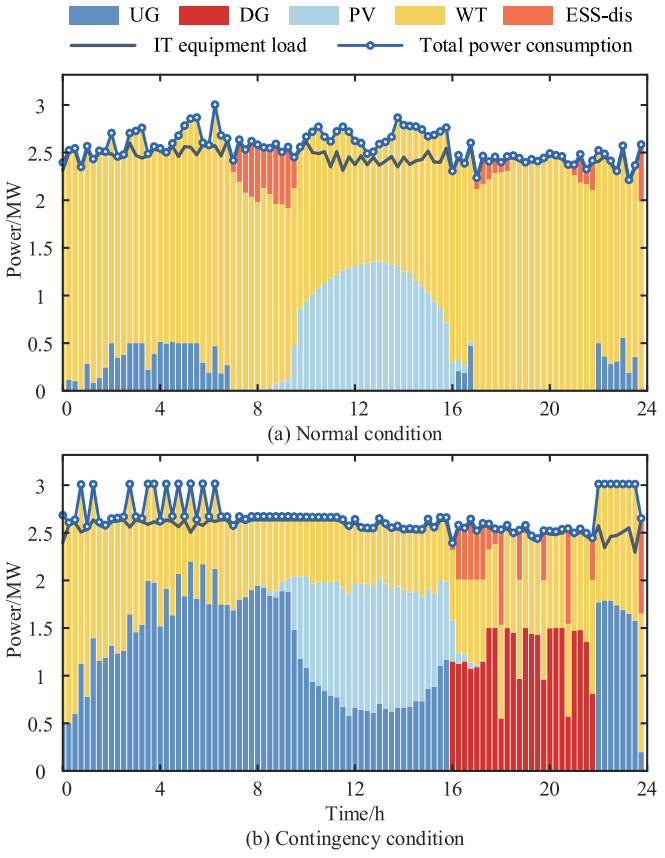


Fig. 8. Representative daily operating profiles under normal and contingency conditions: (a) Normal condition. (b) Contingency condition.

Under the contingency condition in Fig. 8(b), the dispatch pattern changes significantly. During the fault period, the utility grid contribution is reduced, and the DG is activated to provide backup power together with renewable generation and ESS discharging. Although the supply structure changes, the IT load remains largely maintained, which shows that critical computing demand can still be supported under the planned configuration. Meanwhile, the total power consumption does not increase sharply during the outage period, because flexible workload scheduling helps avoid excessive instantaneous demand. This result further explains why Scheme II can achieve a lower investment cost while keeping the reliability-related cost at a limited level.

To further explain the effect of workload flexibility, Fig. 9 compares the delay-tolerant workload profiles before and after scheduling. The two profiles have similar total workload levels, which indicates that the proposed model changes the processing time of delay-tolerant workloads rather than reducing the computing service demand. The scheduled profile fluctuates more significantly, showing that part of the workload is shifted to other intervals according to the operating condition of the data center.

The detailed shifting relationship is shown in Fig. 10. The vertical axis denotes the original arrival time, and the horizontal axis denotes the processing time after scheduling. Each colored element represents the amount of workload shifted from an arrival interval to a processing interval. The diagonal line corresponds to immediate processing, where the processing time is the same as the original arrival time. Most nonzero

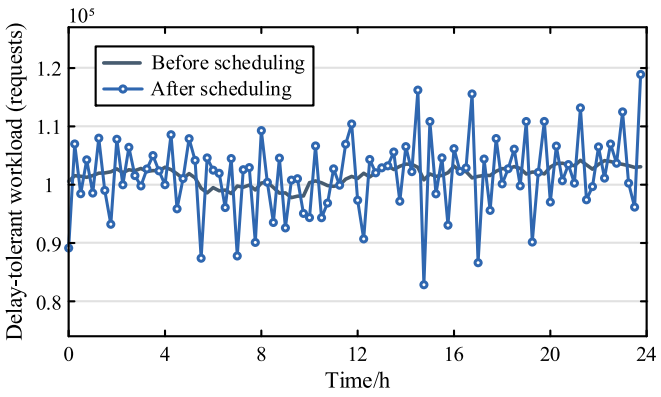


Fig. 9. Scheduling result of delay-tolerant workloads.

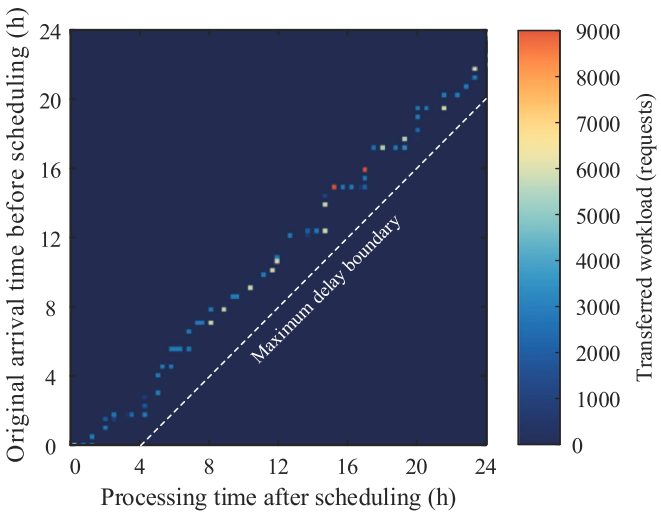


Fig. 10. Workload shifting pattern of delay-tolerant workloads.

elements are distributed on or close to the diagonal, indicating that a large proportion of delay-tolerant workloads are either processed immediately or deferred only for a short period. In addition, all shifted workloads are located within the maximum delay boundary, which confirms that the workload scheduling decisions satisfy the delay-tolerance constraints. The relatively sparse off-diagonal elements further indicate that only part of the workload is shifted to later intervals when beneficial for power-side operation. Therefore, delay-tolerant workloads provide temporal flexibility for data center operation while maintaining the required service completion guarantees.

5.3. Impact of workload service-level composition

To investigate how workload service-level composition affects the coordinated planning of computing and power resources, two sensitivity analyses are conducted. One varies the overall proportion of delay-sensitive and delay-tolerant workloads while keeping the internal composition of delay-tolerant workloads unchanged. This reflects the influence of the total amount of workload available for temporal scheduling. The other fixes the overall proportions of delay-sensitive and delay-tolerant workloads and changes only the internal composition of the delay-tolerant group, so that the effect of delay-tolerance depth can be examined separately. The resulting CPU and GPU deployment, peak IT power, backup cost, and total annual cost are compared.

As shown in Table 5, increasing the share of delay-tolerant workloads substantially changes the optimal server configuration. When the delay-tolerant proportion increases from 20% to 80%, the number of CPU servers decreases from 32,571 to 7908, whereas the number of GPU servers increases from 1298 to 4994. The total computing workload remains unchanged across these cases. However, a larger delay-tolerant share allows more workloads to be shifted within their respective SLO windows and allocated between heterogeneous servers more flexibly.

Under the adopted parameters, GPU servers have a stronger relative processing capability for delay-tolerant workloads, and this advantage increases with delay tolerance. As more workloads become schedulable, a larger share is therefore assigned to GPU servers. Since the total computing demand remains unchanged, the increased GPU deployment replaces part of the required CPU capacity rather than reflecting additional computing demand. Greater schedulable workload also allows computing demand to be redistributed from concentrated intervals to periods with lower system pressure. Consequently, as the delay-tolerant share increases from 20% to 80%, the peak IT power decreases from 2.539 MW to 2.020 MW. Thus, workload flexibility reshapes both heterogeneous server deployment and the temporal profile of IT power demand.

Changes on the computing side further affect backup resource planning. As the peak IT demand declines, less backup support is generally required during utility interruption scenarios. Accordingly, the backup cost decreases from 1.683 million CNY to 1.272 million CNY as the delay-tolerant share increases from 20% to 60%. When the delay-tolerant share reaches 80%, the backup cost increases slightly to 1.311 million CNY. This small increase is associated with the discrete adjustment of modular backup capacities and the corresponding ESS charging and discharging schedules. It does not alter the overall trend that greater workload flexibility reduces the peak-driven pressure placed on the backup supply system.

The total annual cost follows the same broad trend, decreasing from 61.382 million CNY to 41.248 million CNY. This reduction is not caused by a decrease in the total computing workload. Instead, the same workload is served through a different CPU and GPU configuration, a lower peak IT power, and reduced requirements for power conversion, renewable generation, and backup resources. Therefore, increasing the share of delay-tolerant workloads creates planning value through both heterogeneous server reconfiguration and temporal demand redistribution.

The influence of delay-tolerance depth is examined by fixing the proportions of delay-sensitive and delay-tolerant workloads at 40% and 60%, respectively, while varying the internal composition of the delay-tolerant group. As the composition shifts toward workloads with shorter allowable delay windows, a larger fraction of the flexible demand must be completed within a limited period. The total amount of delay-tolerant workload remains unchanged, but the scheduling space available to the model is gradually reduced.

As shown in Table 6, the optimal CPU deployment remains close to 16,000 servers across the four cases, whereas the number of GPU servers increases from 2927 to 4245 as the workload composition shifts from P80-dominated to P95-dominated. Since the overall proportion of delay-sensitive workloads remains unchanged, the computing capacity required for immediate workload processing varies only slightly. This helps explain why the CPU configuration remains broadly stable.

By contrast, shorter delay windows reduce the opportunity to distribute delay-tolerant workloads over time and require more of them to be processed concurrently. In addition, under the adopted parameters, the relative processing capability of GPU servers is lower for P95

Table 5
Planning results under different workload compositions.

Delay-sensitive/ tolerant ratio	Number of servers		Peak IT power (MW)	Backup cost (million CNY)	Total cost (million CNY)
	CPU	GPU			
0.8/0.2	32,571	1298	2.539	1.683	61.382
0.6/0.4	23,145	2927	2.406	1.356	54.530
0.4/0.6	15,820	3839	2.196	1.272	47.808
0.2/0.8	7908	4994	2.020	1.311	41.248

Table 6
Planning results under different delay-tolerant workload compositions.

P95/P90/ P80 ratio	Number of servers		Peak IT power (MW)	Backup cost (million CNY)	Total cost (million CNY)
	CPU	GPU			
0.05/0.2/ 0.35	16,130	2927	1.943	1.215	43.230
0.15/0.2/ 0.25	15,887	3417	2.074	1.258	45.408
0.25/0.2/ 0.15	15,820	3839	2.196	1.272	47.808
0.35/0.2/ 0.05	15,960	4245	2.327	1.502	50.275

workloads than for P80 workloads. Therefore, more GPU servers are required to complete the same amount of delay-tolerant workload within the shorter allowable processing windows.

As a result, the peak IT power increases from 1.943 MW to 2.327 MW because more computing capacity must operate within concentrated intervals. The higher peak IT demand also increases the backup requirement during utility outage scenarios. Accordingly, the backup cost increases from 1.215 million CNY to 1.502 million CNY. The more pronounced increase in the final case also reflects the discrete nature of backup planning, since a further reduction in scheduling flexibility may trigger additional backup capacity or more intensive ESS and DG operation. Correspondingly, the total annual cost rises from 43.230 million CNY to 50.275 million CNY.

The two analyses show that workload flexibility reshapes heterogeneous server deployment through the joint adjustment of workload timing and server allocation. A higher share of delay-tolerant workloads shifts the optimal configuration toward more GPU and fewer CPU servers while reducing peak IT demand, whereas shorter delay windows mainly increase GPU deployment and peak IT demand with little change in CPU capacity. These results indicate that the planning value of workload flexibility arises from the coordinated balance among scheduling flexibility, processing efficiency, and resource cost.

5.4. Sensitivity analysis of workload fluctuation

With the rapid growth of AI workloads, the computing demand of data centers may become more volatile. Compared with conventional cloud services, AI training, inference, and large-model services may introduce more irregular variations and short-duration spikes. However, this volatility depends strongly on the observation scale. Although the utilization of an individual accelerator or server may vary over a wide range, campus-level demand aggregates a large number of heterogeneous tasks and generally includes persistent computing services. The diversity and temporal overlap of these workloads smooth part of the device-level fluctuations, making repeated transitions between zero and full load less representative at the 15 min campus scale considered in this planning study. Nevertheless, stronger aggregate fluctuations may still occur, particularly with the increasing deployment of large-scale AI workloads. To evaluate their potential impact on planning results, three representative workload profiles are considered: baseline, moderate fluctuation, and high fluctuation. The latter two progressively increase the amplitude and frequency of 15 min variations and introduce short-duration spikes spanning one to several planning intervals, while keeping the average workload level and SLO composition unchanged.

Fig. 11 presents representative one-month segments of the constructed workload profiles under different fluctuation levels. Compared with the baseline profile, the profiles under moderate and high fluctuation conditions exhibit more frequent short-term variations and higher local peaks, representing workload conditions with greater temporal variability that may arise from computing services with stronger temporal fluctuations. The one-month segments are shown only for visual clarity, whereas the complete annual profiles with a 15 min resolution are used in all planning and operational optimizations. This temporal resolution is consistent with the annual planning framework and captures the workload variations that materially affect peak demand, resource sizing, and operation under different scenarios.

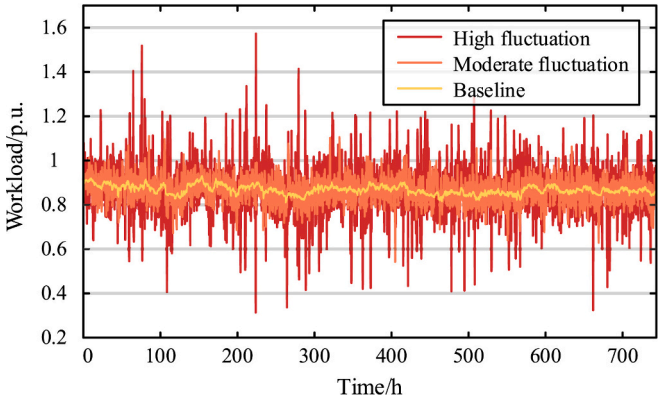


Fig. 11. Aggregated workload profiles of the data center park at 15 min resolution under different fluctuation levels.

The planning results are summarized in Table 7. As workload fluctuation becomes stronger, the peak total power demand increases, which requires additional computing and power resources to satisfy the higher peak requirement. In particular, the capacities of both SST conversion stages increase under the high-fluctuation condition, while the WT capacity and the numbers of CPU and GPU servers also increase as the fluctuation level rises. The additional server capacity is required to process the more concentrated computing demand within the corresponding SLO constraints, whereas the increased SST conversion capacity provides sufficient conversion capability for the higher peak load. These results show that workload fluctuation affects not only the configuration of power conversion and renewable generation resources, but also the deployment of heterogeneous computing resources.

From the cost perspective, stronger workload fluctuations increase the total annual cost primarily through additional investment in servers, WT capacity, and the SST conversion stages. Under the moderate-fluctuation condition, the increased WT capacity provides more local renewable generation and reduces grid electricity purchases, leading to a lower operating cost than in the baseline case. However, under the high-fluctuation condition, the substantially larger peak demand requires considerable expansion of both computing and power conversion capacities. Although the WT capacity is further increased, the additional local generation is insufficient to fully offset the operating requirements associated with the much stronger workload variations and higher peak demand. Consequently, both grid electricity cost and operating cost increase in the high-fluctuation case. The pronounced increase in annualized investment further raises the total annual cost. Overall, the results demonstrate that increasingly severe workload variability can substantially affect computing capacity, conversion capacity, local renewable deployment, and the resulting economic performance, while the proposed planning framework remains applicable under substantially more volatile workload conditions.

5.5. SST loss representations and coefficient variation

To evaluate the influence of SST loss modeling on long-term planning, four loss representations are compared: the lossless, fixed-efficiency, linear, and proposed nonlinear models. Each model is first used to obtain its corresponding capacity configuration. The resulting capacities are then fixed, and the operational problem is re-evaluated using the nonlinear SST loss model. Therefore, the costs in Fig. 12 reflect the performance of different planning configurations under the same loss representation.

As shown in Fig. 12, the proposed nonlinear SST loss model achieves the lowest total annual cost, which is 43.23 million CNY. In contrast, the total costs of the lossless, fixed-efficiency, and linear loss models are 43.83, 43.74, and 43.39 million CNY, respectively. Compared with the proposed nonlinear model, these three simplified models increase the total annual cost by 1.38%, 1.17%, and 0.38%, respectively. This indicates that simplified SST loss representations may lead to suboptimal planning configurations when their results are evaluated under more realistic nonlinear operating conditions.

The cost differences mainly arise from the mismatch between the loss representation used during planning and the actual nonlinear loss characteristics. The lossless model underestimates the required supply margin because conversion losses are neglected, while the fixed-efficiency model cannot reflect the variation of losses with converter loading. The linear model provides a closer approximation but still fails

Table 7
Planning and operating results under different workload fluctuation levels.

Workload profile	Peak total power demand (MW)	SST capacity (MW)		WT capacity (MW)	Number of servers		Grid electricity cost (million CNY)	Investment cost (million CNY)	Operating cost (million CNY)	Total cost (million CNY)
		AC/DC	DC/DC		CPU	GPU				
Baseline	2.635	2.5	3.0	4.5	15,820	3839	6.523	39.333	8.267	47.808
Moderate fluctuation	3.082	2.5	3.5	5.5	17,268	4762	5.344	45.427	7.253	52.685
High fluctuation	4.154	3.0	4.5	7.0	26,184	5136	7.686	64.312	10.032	74.757

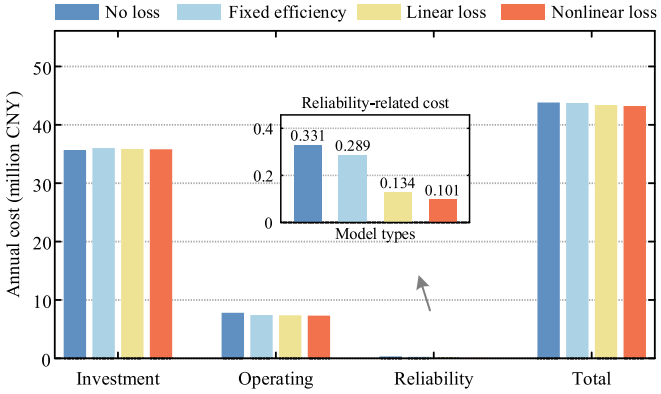


Fig. 12. Annual cost comparison under different SST loss models.

to capture the nonlinear increase in SST losses under high loading. In contrast, the proposed nonlinear model maintains consistency between planning and operation and therefore achieves the best overall tradeoff among investment, operating, and reliability-related costs. When all planning schemes are re-evaluated using the nonlinear SST loss model, the proposed model yields an operating cost of 7.32 million CNY and a reliability-related cost of 0.101 million CNY. As shown in the inset of Fig. 12, the reliability-related cost decreases from 0.331 million CNY in the lossless case to 0.101 million CNY in the proposed nonlinear case.

The larger variation in reliability-related cost arises from the threshold behavior of reliability events. Investment and normal operating costs generally change smoothly with capacity and dispatch decisions, whereas reliability costs can increase sharply once the available supply margin becomes insufficient under contingency conditions. During utility outages or periods of limited supply, SSTs and backup resources are more likely to operate close to their capacity limits, where nonlinear conversion losses become more pronounced. If a simplified loss model underestimates these losses, a configuration that appears economical under normal operation may become insufficient under nonlinear loss evaluation. Consequently, even a moderate loss estimation error can lead to power shortage or load shedding and cause a substantial increase in reliability-related cost. This comparison demonstrates the importance of incorporating nonlinear SST loss characteristics into the planning model for both economic planning and reliability assessment of SST-based DC data centers.

While the above comparison demonstrates the importance of SST loss modeling, the adopted coefficients may vary with converter rating, topology, and device technology. A sensitivity analysis is therefore conducted by uniformly scaling the constant, linear, and quadratic loss coefficients of both the AC/DC and DC/DC stages by 0.75, 1.00, 1.25, and 1.50. This treatment changes the overall loss level while preserving the original dependence of loss on converter loading. For each coefficient setting, the complete planning model is re-optimized to evaluate the resulting changes in resource configuration and scenario operation.

The sensitivity results are summarized in Table 8. As the scaling factor increases from 0.75 to 1.50, the annual SST loss energy rises from 205.63 MWh to 382.74 MWh, while the operating cost increases from 7.204 million CNY to 7.607 million CNY. These results are consistent with the physical meaning of the loss coefficients, because larger coefficients lead to higher conversion losses under similar loading conditions. However, the changes are not proportional to the scaling factor because the capacity configuration and operating schedules are re-optimized in each case.

The capacity results exhibit a modular adjustment pattern. When the scaling factor reaches 1.25, the AC/DC capacity increases from 2.0 MW to 2.5 MW to provide additional conversion margin. When the factor further increases to 1.50, the diesel generator capacity rises from 1.0 MW to 1.5 MW to strengthen backup supply during periods of limited utility support. By contrast, the DC/DC capacity and energy storage capacity remain unchanged in all tested cases. Correspondingly, the total annual cost increases from 43.013 million CNY to 44.269 million CNY as the scaling factor rises from 0.75 to 1.50.

The reliability-related cost does not vary monotonically because it is jointly determined by conversion losses, backup capacity, and dispatch decisions. For example, the additional diesel generator capacity in the 1.50 case reduces load shedding during outages and therefore lowers the reliability-related cost. Overall, the results show that SST loss coefficients affect annual loss energy, operating cost, and selected capacity decisions, while all tested cases remain feasible and yield stable planning solutions. More detailed loss data from manufacturers or specific converter implementations can be incorporated directly by updating the corresponding coefficients or loss curves. The proposed framework can provide a stable basis for evaluating the planning effects of SST losses.

Table 8
Sensitivity results under different SST loss coefficients.

Scaling factor	SST loss energy (MWh)	SST capacity (MW)		ESS capacity (MWh)	DG capacity (MW)	Reliability-related cost (million CNY)	Operating cost (million CNY)	Investment cost (million CNY)	Total cost (million CNY)
		AC/DC	DC/DC						
0.75	205.63	2.0	3.0	3.0	1.0	0.118	7.204	35.691	43.013
1.00	256.98	2.0	3.0	3.0	1.0	0.103	7.318	35.809	43.230
1.25	315.46	2.5	3.0	3.0	1.0	0.117	7.453	36.082	43.652
1.50	382.74	2.5	3.0	3.0	1.5	0.068	7.607	36.594	44.269

5.6. Backup resource coordination under outage conditions

Backup resources play a critical role in maintaining data center service continuity during utility supply interruptions. In this subsection, four backup configuration cases are compared under different operating conditions. Case I does not include dedicated backup resources, Case II only considers ESS, Case III only considers DG, and Case IV jointly considers ESS and DG. Four operating conditions are examined, including normal operation, short outage, long outage, and frequent outage. The normal operating condition assumes that the utility grid is continuously available. The short outage condition introduces a temporary utility interruption over a limited outage window, while the long outage condition extends the outage duration to represent sustained supply loss. The frequent outage condition contains several outage windows within the scheduling horizon, so as to represent repeated utility disturbances. These settings are used to distinguish the effects of outage duration and occurrence frequency on backup resource configuration.

To compare the economic effects of different backup configurations, Fig. 13(a) reports the backup-related cost, which includes the annualized investment cost of ESS and DG, the operating cost of ESS and DG, and the reliability-related loss cost caused by insufficient supply during outage conditions. Under normal operation, Case I has nearly zero backup-related cost because no dedicated backup resources are deployed and no reliability-related loss is incurred. However, under outage conditions, although Case I still has no backup investment or backup operating cost, its reliability-related loss cost increases rapidly due to the lack of emergency power support. The comparison between Cases II and III reflects the different operating characteristics of ESS and DG. Under short outage conditions, Case II shows better performance because ESS can provide fast power support for short-duration disturbances. However, under long and frequent outage conditions, the cost of Case II increases more significantly due to the limited energy duration of storage. In contrast, DG can provide sustained backup power, making Case III more suitable for long-duration outages. Nevertheless, relying only on DG lacks the fast response and flexible regulation capability of ESS.

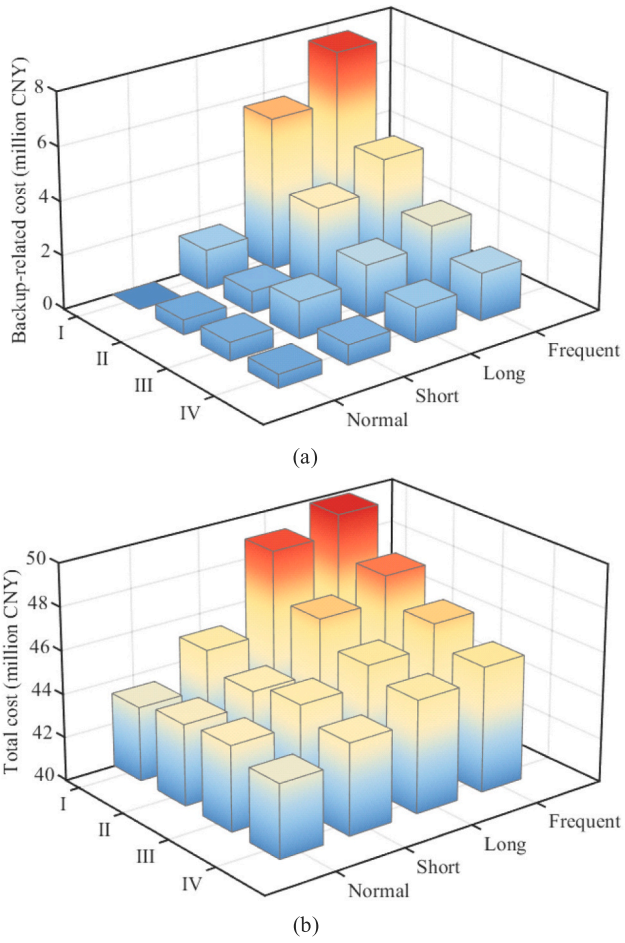


Fig. 13. Impact of backup resource configuration under different operating conditions: (a) Backup-related annual cost; (b) Total annual cost.

Fig. 13(b) compares the total annual cost of different backup configurations. Case IV achieves the lowest total cost under outage conditions, and its advantage becomes more evident as the outage severity increases. This is because the coordinated ESS–DG configuration can adapt to different outage patterns. For short outages, ESS provides rapid power support, while for long and frequent outages, DG provides sustained backup supply. Therefore, the coordinated configuration achieves a better balance among backup investment, operating cost, and reliability-related losses.

These results indicate that the value of backup resources is closely related to the characteristics of supply interruptions. ESS is more effective for short-term disturbances, whereas DG is more suitable for long-duration outages. Compared with single-resource configurations, the coordinated ESS–DG configuration provides a more flexible and robust backup structure, thereby improving the economic performance of SST-based DC data centers under severe outage conditions.

5.7. Impact of PUE on energy consumption and planning

Power usage effectiveness (PUE) directly affects the auxiliary electricity consumption associated with cooling, power distribution, and other supporting facilities in data centers. In the proposed model, auxiliary power consumption is represented as a fixed proportion of IT power consumption through the PUE parameter. Although this treatment provides a tractable planning-level representation, the actual PUE of a data center may vary with cooling-system efficiency, outdoor meteorological conditions, equipment loading, and operating strategies. Therefore, a sensitivity analysis is conducted to examine how different aggregate auxiliary energy efficiency levels affect annual energy consumption, resource configuration, and total planning cost. Four PUE values, namely 1.1, 1.2, 1.3, and 1.4, are considered. For all cases, the workload profile, workload composition, renewable-generation profiles, electricity tariffs, utility-outage settings, and equipment parameters are kept unchanged, and the complete planning model is re-optimized for each PUE value. The comparison focuses on annual IT and auxiliary energy consumption, utility-purchased energy, renewable energy output, peak total load, resource capacities, and total annual cost.

As shown in Table 9, annual IT energy consumption remains relatively stable across the four cases because the workload profile, workload composition, and service requirements are unchanged. The small variation in IT energy consumption results from the re-optimized workload scheduling, server operation, and DVFS decisions rather than from any reduction in computing demand. In contrast, annual auxiliary energy consumption increases markedly from 1.843 GWh to 7.324 GWh as the PUE increases from 1.1 to 1.4. Since auxiliary power is coupled with IT power in the present formulation, a higher PUE increases the total electricity demand throughout the scheduling horizon. Consequently, the peak total load increases by approximately 26.4%. This result indicates that auxiliary energy efficiency directly affects both annual energy consumption and the maximum power requirement that must be supported by the data center power supply system.

The additional auxiliary demand also changes the optimal balance between utility purchases and on-site renewable generation. Although workload flexibility can shift part of the IT demand to more favorable periods, it cannot eliminate the auxiliary energy associated with processing the required workloads. Therefore, as the PUE increases, the system needs to obtain more energy from both the utility grid and local

Table 9
Impact of PUE on energy consumption and planning results.

PUE	Annual IT energy (GWh)	Annual auxiliary energy (GWh)	Utility-purchased energy (GWh)	Renewable energy output (GWh)	Peak total load (MW)	SST capacity (MW)		PV capacity (MW)	WT capacity (MW)	Total cost (million CNY)
						AC/DC	DC/DC			
1.1	18.435	1.843	6.770	13.617	2.443	2.5	3.0	1.5	4.5	46.806
1.2	18.231	3.646	8.112	13.859	2.635	2.5	3.0	1.5	4.5	47.808
1.3	18.213	5.464	8.723	15.012	2.870	2.5	3.0	1.5	5.0	48.920
1.4	18.311	7.324	8.858	16.895	3.088	3.0	3.5	2.0	5.5	50.028

renewable resources. The gradual increase in renewable output and optimized PV and WT capacities show that PUE affects not only short-term electricity purchasing decisions, but also the long-term economic sizing of renewable generation.

The capacity results further show that the planning response to PUE is not fully continuous. The AC/DC and DC/DC capacities remain unchanged as the PUE increases from 1.1 to 1.3, indicating that the original SST configuration retains sufficient operating margin under moderate increases in auxiliary demand. When the PUE reaches 1.4, however, the increase in peak load exceeds this margin and triggers the installation of additional SST modules. This stepwise variation is consistent with the modular capacity constraints adopted in the planning model. Meanwhile, the total annual cost increases from 46.806 million CNY to 50.028 million CNY because higher PUE values increase electricity purchases and renewable supply requirements and may eventually require additional power conversion capacity. Overall, the results confirm that auxiliary energy efficiency has a material influence on both operating energy requirements and long-term resource configuration, while the proposed planning framework can consistently capture these effects over the considered PUE range.

5.8. Comparison of SST and UPS architectures

To provide a benchmark for the proposed SST-based DC supply system, a conventional architecture incorporating a line-frequency transformer and an uninterruptible power supply (UPS) is further considered. To isolate the effects of the power supply architecture, the installed CPU and GPU quantities and the resulting data center load profile are fixed at the values obtained in the SST baseline case. The installed PV and WT capacities and their corresponding output profiles are also kept unchanged. In addition, the same electricity prices, utility outage scenarios, DG and ESS parameters, and reliability criteria are adopted for both architectures. Under these common boundary conditions, the UPS conversion capacity, DG capacity, ESS capacity, grid purchases, and operating schedules are re-optimized. The conventional supply chain is represented by an equivalent conversion efficiency of 0.93, selected with reference to the UPS efficiency characteristics reported in [48]. The unit investment costs of the SST and UPS conversion equipment are set to 1000 CNY/kW and 600 CNY/kW, respectively. For the SST architecture, the AC/DC and DC/DC stages are independently sized, and the capacities of both stages are included in the conversion equipment investment. Only the architecture-dependent power conversion, backup resource, operating, and reliability-related costs are included in this comparison.

As shown in Table 10, the SST architecture installs 2.5 MW of AC/DC capacity and 3.0 MW of DC/DC capacity, with both stages included in the investment calculation. Its initial conversion equipment investment is therefore 5.5 million CNY, compared with 1.8 million CNY for the 3.0 MW UPS. After including DG and ESS, the total architecture-dependent initial investments are 11.5 million CNY for the SST architecture and 8.8 million CNY for the UPS architecture, corresponding to annualized investment costs of 1.171 million CNY and 0.896 million CNY, respectively. Thus, the UPS architecture has an annualized investment advantage of 0.275 million CNY.

However, the higher conversion losses of the UPS architecture increase its operating and backup requirements. Under the same computing load and renewable generation conditions, annual conversion losses increase from 0.257 GWh to 1.653 GWh, increasing utility-purchased energy from 8.115 GWh to 9.550 GWh and operating cost from 7.831 million CNY to 8.564 million CNY. During utility outages, the same computing load must be maintained under the same renewable generation level. The higher conversion losses of the UPS architecture therefore increase the net power demand that must be supplied by on-site backup resources, leading to greater backup capacity requirements and higher reliability-related costs. As shown in Table 10, the optimized DG capacity increases from 1.5 MW to 2.0 MW, corresponding to an additional initial DG investment of 1.0 million CNY and an annualized cost of 0.102 million CNY. Meanwhile, the reliability-related cost increases from 0.208 million CNY to 0.236 million CNY.

Overall, the SST architecture reduces the combined annual operating and reliability-related costs by 0.761 million CNY. After accounting for the difference in total architecture-dependent initial investment, the corresponding simple payback period is approximately 4 years under the adopted load, electricity price, outage, and equipment cost assumptions. The resulting total annual cost is 9.210 million CNY for the SST architecture and 9.696 million CNY for the UPS architecture, representing an annual saving of 0.486 million CNY. These results indicate that the higher initial investment of the SST architecture can be recovered through lower conversion losses, reduced grid electricity purchases, and reduced backup requirements. Nevertheless, the economic comparison remains dependent on equipment prices, conversion characteristics, project scale, electricity tariffs, and reliability requirements. Further reductions in SST costs resulting from advances in semiconductor devices, standardized equipment design, and manufacturing scale may further improve its economic competitiveness.

5.9. Computational performance analysis

The proposed coordinated planning model contains a large number of scenario-dependent operating variables and several types of integer variables, including modular capacity decisions, DG commitment states, ESS charging/discharging status variables, and DVFS frequency

Table 10
Comparison of planning and operating results between SST and UPS architectures.

Architecture	Power conversion capacity (MW)	DG capacity (MW)	ESS capacity (MWh)	Annual conversion loss (GWh)	Utility-purchased energy (GWh)	Investment cost (million CNY)	Operating cost (million CNY)	Reliability cost (million CNY)	Total cost (million CNY)
SST	2.5/3.0	1.5	2.0	0.257	8.115	1.171	7.831	0.208	9.210
UPS	3.0	2.0	2.0	1.653	9.550	0.896	8.564	0.236	9.696

selection variables. In addition, the nonlinear SST loss characteristics are represented by piecewise linear approximation, which further increases the model size. Therefore, it is necessary to evaluate the computational performance of the proposed enhanced Benders decomposition method.

Fig. 14 illustrates the iterative convergence of the proposed enhanced Benders decomposition method. The upper bound corresponds to the objective value of the best feasible planning solution obtained at each iteration, whereas the lower bound represents the master problem's objective value. The relative gap, defined as the normalized difference between the upper and lower bounds, decreases monotonically as the iterations progress. In the initial stage, the relative gap drops sharply, reflecting rapid improvement in the incumbent feasible solution through scenario subproblem evaluations. Concurrently, the master problem is incrementally tightened by the accumulated unified Benders cuts and feasibility restoration cuts, leading to a steady increase in the lower bound. As the iterations continue, the reduction of the relative gap slows and gradually stabilizes near the predefined convergence tolerance of 0.1%, indicating that the algorithm has effectively closed the optimality gap. Overall, Fig. 14 demonstrates that the proposed method reliably achieves convergence within 47 iterations, balancing computational efficiency and solution quality for large-scale multi-scenario planning problems.

To further demonstrate the computational advantage of the proposed method, several benchmark methods are compared in Table 11. In the table, EBD-I denotes the proposed enhanced Benders decomposition method with parallel subproblem solution and feasibility restoration cuts. EBD-II denotes the sequential version of the proposed method, in which the daily scenario subproblems are solved one by one without parallelization. EBD-III denotes the proposed method without feasibility restoration cuts. In addition, two representative evolutionary algorithms, namely differential evolution (DE) and the nondominated sorting genetic algorithm III (NSGA-III), are included for comparison.

As shown in Table 11, EBD-I converges within 47 iterations and obtains a best total cost of 43.077 million CNY with a final gap of 0.08%. In the annual implementation, the 35,040 scheduling intervals are organized into 365 daily scenarios, with each scenario containing 96 intervals. After the first-stage planning decisions are fixed, the corresponding daily operational subproblem is formulated as a MILP, since the operational objective and constraints are linear under the piecewise linear representation of SST losses while the operational binary decisions are retained. Under the case study configuration, each daily subproblem contains 33,216 variables, including 9024 binary variables, and 62,274 constraints. The cut-generation procedure follows Sections 4.3 and 4.4, where the incumbent binary recourse decisions are fixed to obtain a linear representation for deriving the Benders cuts from dual information. EBD-I and EBD-II employ the same daily subproblem formulation and therefore have identical individual subproblem sizes. The two methods differ in the solution strategy for the daily scenarios:

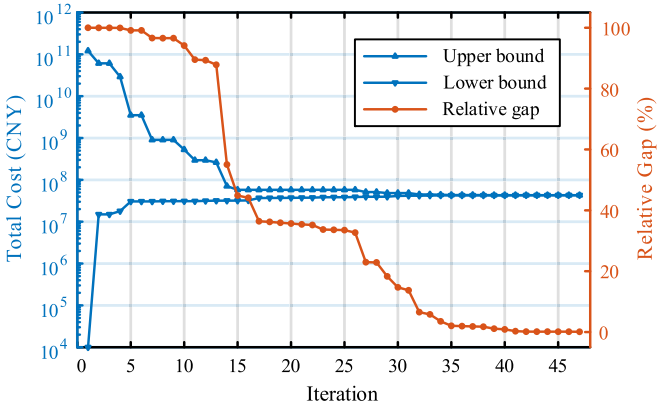


Fig. 14. The enhanced Benders decomposition iterative solution process.

Table 11
Computational performance comparison of different methods.

Method	Convergence status	Iterations or evaluations	Runtime (h)	Total cost (million CNY)	Final gap
EBD-I	Converged	47	4.29	43.077	0.08%
EBD-II	Converged	47	58.48	43.077	0.08%
EBD-III	Not converged	80	9.85	43.513	0.61%
DE	Terminated	600	46.73	62.359	/
NSGA-III	Terminated	600	51.28	56.742	/

EBD-II solves the subproblems sequentially, whereas EBD-I evaluates multiple independent subproblems concurrently under a parallel computing configuration with up to 16 solver threads in aggregate. Consequently, the total runtime decreases from 58.48 h for EBD-II to 4.29 h for EBD-I, while both methods reach the same total cost and final convergence gap. These results indicate that parallel scenario evaluation substantially improves the computational efficiency of the annual multi-scenario planning problem.

The comparison between EBD-I and EBD-III shows the effect of feasibility restoration cuts on convergence behavior. Without these cuts, EBD-III still obtains a feasible solution with a total cost of 43.5126 million CNY, but it does not reach the prescribed convergence tolerance within 80 iterations. Its final gap remains 0.61%, indicating that the unified Benders cuts alone may not remove unfavorable master solutions sufficiently fast when binary recourse variables are present. By adding feasibility restoration cuts, EBD-I improves the iterative refinement of the master problem and achieves more stable convergence.

DE and NSGA-III are included as heuristic benchmarks to examine the benefit of exploiting the problem structure. Under the adopted evaluation limit, both methods obtain feasible solutions but do not provide rigorous lower bounds or optimality gaps. Therefore, their results are reported only as feasible solution costs rather than strict convergence gaps. As shown in Table 11, DE and NSGA-III terminate after 600 evaluations, with runtimes of 46.73 h and 51.28 h, respectively. Their total costs are 62.3587 million CNY and 56.7423 million CNY, which are much higher than that of EBD-I. This indicates that the heuristic methods can easily become trapped in local optima or inefficient feasible regions for this large-scale mixed-integer planning problem. In contrast, the decomposition-based method explicitly exploits the scenario separability and mathematical structure of the model, leading to a more reliable planning solution.

In summary, the computational results show that the proposed EBD-I method provides a practical solution approach for the annual coordinated planning problem. It achieves a high-quality solution within an acceptable runtime, while the comparisons with EBD-II and EBD-III confirm the benefits of parallel subproblem solution and feasibility restoration cuts. The heuristic benchmarks further indicate that exploiting the decomposable structure of the proposed model is important for obtaining reliable planning results efficiently.

6. Conclusion

This paper develops a coordinated planning and operation framework for SST-based DC data centers with tiered workload flexibility. It integrates power infrastructure planning, computing resource allocation, and scenario-dependent operation within a unified annual framework, while explicitly representing nonlinear SST losses and service level constraints. A piecewise linear reformulation and an enhanced Benders decomposition method are used to improve the tractability of the resulting large-scale multi-scenario problem. The case studies demonstrate that workload flexibility can reshape temporal demand and heterogeneous server deployment, thereby reducing peak power requirements, conversion capacity, backup needs, and annual cost. The

results also show that simplified SST loss models may distort capacity decisions and reliability assessment under stressed conditions. Further analyses confirm that workload fluctuation, service level composition, PUE, and outage characteristics materially influence the coordinated planning results. Overall, the proposed framework provides a practical tool for evaluating the long-term interaction between computing flexibility and power resource configuration in SST-based DC data centers.

Future work will extend the proposed framework by developing an integrated planning model for electricity, cooling, and computing. Outdoor temperature, cooling modes, power usage effectiveness, and the coefficient of performance (COP) of cooling equipment will be incorporated to characterize the temporal interaction between workload scheduling and cooling demand. Alternative topologies of the power conversion stages will also be considered to evaluate the effects of topology-dependent conversion characteristics on capacity planning and system operation. In addition, the interaction between SST-based DC data centers and upstream distribution networks will be investigated under network constraints and demand response signals.

CRediT authorship contribution statement

Jian Xu: Supervision, Software, Resources, Methodology, Investigation. **Qiang Xu:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Xinxiong Jiang:** Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Conceptualization. **Siyang Liao:** Project administration, Methodology, Investigation. **Deping Ke:** Supervision, Software, Resources. **Liangzhong Yao:** Visualization, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendices

Appendix A

Parameters of the Data Center

Parameter	Value
Frequency levels of CPU servers (GHz)	2.0/2.4/2.8/3.2/3.6
Frequency levels of GPU servers (GHz)	1.1/1.3/1.5/1.7
Fixed power consumption of CPU/GPU (kW/server)	0.053/0.250
Power consumption coefficient κ_1 of CPU/GPU (kW/GHz^2)	0.013/0.050
Computing efficiency coefficient κ_2 of servers	5
CPU relative processing capability coefficient $\omega_{k,\rho}$	1.0/0.95/0.90/0.85
GPU relative processing capability coefficient $\omega_{k,\rho}$	2.5/2.8/3.2/3.6
CPU/GPU server unit investment cost (CNY/server)	5000/15000
PUE indicator	1.2
Maximum stored workload (requests)	1×10^9

Appendix B

Parameters of Power Supply Equipment

Parameters	Value
SST AC/DC and DC/DC unit capacity	0.5 MW
SST unit capacity investment	1000 CNY/kW
UPS unit capacity	0.5 MW
UPS unit investment cost	600 CNY/kW
Conversion efficiency of UPS architecture	0.93
ESS unit capacity	0.5 MWh
ESS unit investment cost	1500 CNY/kWh
PV unit capacity	0.5 MW
PV unit investment cost	3200 CNY/kW
WT unit capacity	0.5 MW
WT unit investment cost	6500 CNY/kW
DG unit capacity	0.5 MW
DG unit investment cost	2000 CNY/kW
DG operating cost	1.5 CNY/kWh

Appendix C

Settings of utility outage scenarios

Scenario	Outage event setting		Starting time
	Number	Duration	
Normal operation	/	/	/
Short outage	3 events	2 h	Random sampled
Long outage	3 events	8 h	Random sampled
Frequent outage	9 events	6 h	Random sampled

Data availability

Data will be made available on request.

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